

US BANK NA.

By: /s/ Greg A Ferrence
Regional President

FIRSTMERIT BANK, N.A.

By: /s/ Andrea K. Poliak
Assistant Vice President

NUWCDIVNPT VISITOR REQUEST INFORMATION NEEDED:

****Information must be on Company Letterhead****

- 1) Company/Organization Name, Address, Telephone, and FAX Number
- 2) Cage Code
- 3) Visitor(s) Full Name (As it appears on government issued ID – i.e. Driver's License)
- 4) Visitor(s) Full Social Security Number
- 5) Visitor(s) Date of Birth
- 6) Visitor(s) Place of Birth (City & State)
 - a. If born outside the U.S, you MUST provide a Naturalization Number or US Passport
- 7) Visitor(s) Citizenship
- 8) Level of Clearance: UNCLASSIFIED or as appropriate
 - a. SCI Level Access MUST be coordinated SSO to SSO
- 9) Do they possess a CAC, Teslin or DBIDs Card? (Yes or No)
 - a. Please specify which visitor(s) have CAC, Retired Military/Dependent ID, DBIDS, or NOT APPLICABLE
- 10) NUWCDIVNPT Point of Contact (Sponsor)
 - a. MUST be a NUWC Government Employee
- 11) Specific Date(s) of Visit, to include Year
- 12) Purpose of Visit
- 13) Contract Number, if applicable
 - a. If NO Contract Number is provided, then the visit will be for 30 days at the UNCLASSIFIED Level
- 14) Authorized Signature:
 - a. Name, Title, Signature, and Date of Requesting Company/Organization Authorizing Official (MUST be a US Citizen)
 - b. Smaller companies that do not have a Security Officer to sign may use another Official or Secretary.
 - c. Individuals MUST NOT Sign their own request

Submit Visitor Request(s) and/or SECNAV form(s) 5512/1 via:

- **JPAS SMO CODE:** 666045 (If no access to JPAS, use methods below)
- **EMAIL FORM TO:** nuwc_npt_access_control@navy.mil
- ☐ **FAX FORM TO:** (401) 832-4396

****ALL INCOMPLETE VISIT REQUESTS WILL BE DENIED****

Please call NUWC visitor control at (401) 832-2152 if you want to verify any visits or status on vetting.

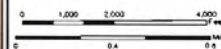
NAVSTA Newport

Commander Navy Region
Mid-Atlantic GeoReadiness
Center

Legend

- Gates
- Base Boundary
- Existing Structures
- Ammunition Storage Area
- Fence Line
- Runway
- Taxiway
- Helipad
- Apron
- Shoulder Overrun
- Aircraft Parking Area
- Railroad
- Golf Course
- Playground
- Swimming Pool
- Athletic Court
- Athletic Field
- Existing Piers
- Drydock

Print Date: 10 Oct, 2021



Coordinate System: NAD83 for US Zone 17 N

GeoReadiness Center

AM-OS Mid-Atlantic
Norfolk, VA 23511
(757) 444-3013



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Published: 10 Oct 2021

INSTRUCTIONS (*Back Page*):

Instructions for items other than those that are self-explanatory, are as follows:

- (a) Item 1 (Contract ID Code). Insert the contract type identification code that appears in the title block of the contract being modified.
- (b) Item 3 (Effective date).
- (1) For a solicitation amendment, change order, or administrative change, the effective date shall be the issue date of the amendment, change order, or administrative change.
- (2) For a supplemental agreement, the effective date shall be the date agreed to by the contracting parties.
- (3) For a modification issued as an initial or confirming notice of termination for the convenience of the Government, the effective date and the modification number of the confirming notice shall be the same as the effective date and modification number of the initial notice.
- (4) For a modification converting a termination for default to a termination for the convenience of the Government, the effective date shall be the same as the effective date of the termination for default.
- (5) For a modification confirming the contracting officer's determination of the amount due in settlement of a contract termination, the effective date shall be the same as the effective date of the initial decision.
- (c) Item 6 (Issued By). Insert the name and address of the issuing office. If applicable, insert the appropriate issuing office code in the code block.
- (d) Item 8 (Name and Address of Contractor). For modifications to a contract or order, enter the contractor's name, address, and code as shown in the original contract or order, unless changed by this or a previous modification.
- (e) Items 9, (Amendment of Solicitation Number - Dated), and 10, (Modification of Contract/Order Number - Dated). Check the appropriate box and in the corresponding blanks insert the number and date of the original solicitation, contract, or order.
- (f) Item 12 (Accounting and Appropriation Data). When appropriate, indicate the impact of the modification on each affected accounting classification by inserting one of the following entries:
- (1) Accounting classification _____
Net increase \$ _____
- (2) Accounting classification _____
Net decrease \$ _____
- NOTE: If there are changes to multiple accounting classifications that cannot be placed in block 12, insert an asterisk and the words "See continuation sheet".
- (g) Item 13. Check the appropriate box to indicate the type of modification. Insert in the corresponding blank the authority under which the modification is issued. Check whether or not contractor must sign this document. (See FAR 43.103.)
- (h) Item 14 (Description of Amendment/Modification).
- (1) Organize amendments or modifications under the appropriate Uniform Contract Format (UCF) section headings from the applicable solicitation or contract. The UCF table of contents, however, shall not be set forth in this document.
- (2) Indicate the impact of the modification on the overall total contract price by inserting one of the following entries:
- (i) Total contract price increased by \$ _____
- (ii) Total contract price decreased by \$ _____
- (iii) Total contract price unchanged.
- (3) State reason for modification.
- (4) When removing, reinstating, or adding funds, identify the contract items and accounting classifications.
- (5) When the SF 30 is used to reflect a determination by the contracting officer of the amount due in settlement of a contract terminated for the convenience of the Government, the entry in Item 14 of the modification may be limited to --
- (i) A reference to the letter determination; and
- (ii) A statement of the net amount determined to be due in settlement of the contract.
- (6) Include subject matter or short title of solicitation/contract where feasible.
- (i) Item 16B. The contracting officer's signature is not required on solicitation amendments. The contracting officer's signature is normally affixed last on supplemental agreements.

AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT				1. CONTRACT ID CODE		PAGE OF PAGES			
2. AMENDMENT/MODIFICATION NUMBER			3. EFFECTIVE DATE		4. REQUISITION/PURCHASE REQUISITION NUMBER		5. PROJECT NUMBER (If applicable)		
6. ISSUED BY CODE					7. ADMINISTERED BY (If other than Item 6) CODE				
8. NAME AND ADDRESS OF CONTRACTOR (Number, street, county, State and ZIP Code)						(X)		9A. AMENDMENT OF SOLICITATION NUMBER	
						☐		9B. DATED (SEE ITEM 11)	
						☐		10A. MODIFICATION OF CONTRACT/ORDER NUMBER	
						☐		10B. DATED (SEE ITEM 13)	
CODE			FACILITY CODE						
11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS									
☐ The above numbered solicitation is amended as set forth in Item 14. The hour and date specified for receipt of Offers ☐ is extended. ☐ is not extended.									
Offers must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended, by one of the following methods: (a) By completing items 8 and 15, and returning _____ copies of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted; or (c) By separate letter or electronic communication which includes a reference to the solicitation and amendment numbers. FAILURE OF YOUR ACKNOWLEDGMENT TO BE RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER. If by virtue of this amendment you desire to change an offer already submitted, such change may be made by letter or electronic communication, provided each letter or electronic communication makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.									
12. ACCOUNTING AND APPROPRIATION DATA (If required)									
13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACTS/ORDERS. IT MODIFIES THE CONTRACT/ORDER NUMBER AS DESCRIBED IN ITEM 14.									
CHECK ONE	A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: (Specify authority) THE CHANGES SET FORTH IN ITEM 14 ARE MADE IN THE CONTRACT ORDER NUMBER IN ITEM 10A.								
☐									
☐	B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT THE ADMINISTRATIVE CHANGES (such as changes in paying office, appropriation data, etc.) SET FORTH IN ITEM 14, PURSUANT TO THE AUTHORITY OF FAR 43.103(b).								
☐	C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:								
☐	D. OTHER (Specify type of modification and authority)								
E. IMPORTANT: Contractor ☐ is not ☐ is required to sign this document and return _____ copies to the issuing office.									
14. DESCRIPTION OF AMENDMENT/MODIFICATION (Organized by UCF section headings, including solicitation/contract subject matter where feasible.)									
15A. NAME AND TITLE OF SIGNER (Type or print)					16A. NAME AND TITLE OF CONTRACTING OFFICER (Type or print)				
15B. CONTRACTOR/OFFEROR			15C. DATE SIGNED		16B. UNITED STATES OF AMERICA		16C. DATE SIGNED		
(Signature of person authorized to sign)					(Signature of Contracting Officer)				

pre-tax charge of \$13.4 million (\$8.1 million after-tax) principally related to the costs associated with the lapsed Charter offer, offset by net proceeds received in settlement of a dispute with one of the Company's product liability insurance carriers.

Interest Expense. Interest expense decreased \$1.9 million, or 25.7% in 2001 from \$7.4 million in 2000. The decrease in interest expense was commensurate with decreased short- and long-term borrowings.

Income Taxes. Income taxes for 2001 were \$32.4 million on income before income taxes of \$116.0 million, an effective rate of 28.0%, as compared with income taxes of \$43.6 million on income before income taxes of \$121.7 million, or an effective rate of 35.8% for 2000. The decrease in the effective tax rate was primarily attributable to the utilization of certain international tax credits.

Net Income. Net income for 2001 of \$83.6 million was \$5.5 million higher than 2000. Diluted earnings per share for 2001 increased to \$1.96 per share from \$1.83 per share in 2000. Net income in 2001 included an after-tax gain of \$1.2 million, without which net income would have been \$82.3 million or \$1.93 per share (diluted). The net income in 2000 included the charges mentioned above. Excluding the nonrecurring items in 2000, net income would have been \$86.2 million or \$2.02 per share (diluted). The effect of foreign currency exchange rate movements on net income was not significant.

LIQUIDITY AND CAPITAL RESOURCES

The Company's debt levels increased from \$50.8 million at December 31, 2001, to \$187.5 million at December 31, 2002. Total percent of debt to total capitalization increased to 30.5% at December 31, 2002, from 9.2% at December 31, 2001. The increase in 2002 is attributable to the \$150 million of Senior Unsecured Notes discussed below. Management anticipates that the Company will be able to satisfy cash requirements for its ongoing businesses for the foreseeable future primarily with existing cash balances, cash generated by operations and, if necessary, borrowings under its existing credit facilities.

Cash provided from operations was \$103.6 million in 2002, a decrease of \$16.6 million from \$120.2 million in 2001. The decrease was primarily related to lower net income.

Capital expenditures during 2002 were \$27.9 million, a 24.0% decrease from 2001. The Company anticipates capital expenditures should approximate depreciation in 2003. If acquisitions and major projects providing near term financial benefits become available, additional expenditures may be made.

During January 2002, the Company acquired 85% of Bester S.A., a leading manufacturer of welding equipment located in Poland, for \$7.8 million, including assumed debt. During 2002, the Company also acquired an additional 6% of Bester through additional investments totaling \$3.5 million, of which \$1.5 million has been paid as of December 31, 2002.

During December 2001, the Company acquired 100% of Messer Soldaduras de Venezuela S.A., a leading manufacturer of consumable welding products located in Venezuela, for \$3.4 million, including assumed debt.

During the first quarter of 2000, the Company acquired a 35% interest in Kuang Tai, a Taiwan-based manufacturer of welding wire for \$16.8 million and 100% of C.I.F.E. S.r.l., an Italian-based manufacturer of MIG wire for \$19.5 million, including assumed debt.

During March 2002, the Company issued Notes totaling \$150 million through a private placement. Maturities and interest rates on the Notes are \$40 million at 5.58% in 2007, \$30 million at 5.89% in 2009 and \$80 million at 6.36% in 2012. Interest is payable semi-annually in March and September. The proceeds will be used for general corporate purposes. The Notes contain certain affirmative and negative

covenants, including restrictions on asset dispositions and financial covenants (interest coverage and funded Debt-to-EBITDA ratios). The Notes will not be and have not been registered under the Securities Act of 1933, as amended, and may not be offered or sold in the United States absent registration or an applicable exemption from registration requirements.

During March 2002, the Company entered into floating rate interest rate swap agreements with notional amounts totaling \$80 million, to convert a portion of the outstanding Notes from fixed to floating rates based on the six-month LIBOR, plus a spread of between 15.5 and 37.5 basis points. The variable rates are reset every six months, at which time payment or receipt of interest is settled. These swaps were designated as fair value hedges, and as such, the gain or loss on the derivative instrument as well as the offsetting gain or loss on the hedged item attributable to the hedged risk are recognized in earnings. Net payments or receipts under these agreements are recognized as adjustments to interest expense. The fair value of these swaps at December 31, 2002 was \$9.9 million. The weighted average effective interest rate on the Notes after considering the effect of the interest rate swaps was 4.82% for 2002. Net interest expense related to the \$150 million private placement is further reduced by the interest income earned on the cash balances. These short-term, highly liquid investments earned 2.23%, or approximately \$2.5 million during 2002. Total net interest expense related to the \$150 million private placement after considering the fixed-to-floating interest rate swaps and the interest income generated from investing the funds was approximately \$2.9 million, or \$0.04 per diluted share.

During April 2002, the Company replaced its prior committed revolving credit facility with a new three-year revolving Credit Agreement that provides up to \$125 million in borrowings and expires in April 2005. This Credit Agreement may be used for general corporate purposes. The interest rate on borrowings under the Credit Agreement is based on either the LIBOR plus a spread based on the Company’s leverage ratio or the prime rate, at the Company’s election. A facility fee is payable based upon the daily aggregate amount of commitments. The facility fee is based on the Company’s leverage ratio. The Credit Agreement provides for the issuance of Letters of Credit, subject to limits based on amounts outstanding under the Credit Agreement and is subject to the same covenants as the Notes. As of December 31, 2002, there are no borrowings under the credit agreement.

During April 2002, the Company amended and restated its existing 8.73% Senior Notes due in 2003. The 8.73% Senior Notes were amended and restated so that the affirmative and negative covenants are consistent with the Notes. In addition, the amendment and restatement of the 8.73% Senior Notes added an uncommitted private shelf facility allowing for the issuance of an aggregate of \$100 million of additional senior unsecured notes (the “Shelf Notes”). There were no borrowings under the Shelf Notes as of December 31, 2002.

The Company’s contractual obligations and commercial commitments (as defined by Section 13(j) of the Securities Exchange Act of 1934) as of December 31, 2002 are as follows (in thousands):

	Total	Payments Due by Period			
		2003	2004 to 2005	2006 to 2007	2008 and Beyond
Long-term debt	\$181,590	\$12,097	\$ 4,028	\$46,913	\$118,552
Capital lease obligations	5,989	1,336	1,611	1,031	2,011
Short-term debt	945	945	—	—	—
Operating leases	20,031	8,068	8,404	3,433	126
Total contractual cash obligations	\$208,555	\$22,446	\$14,043	\$51,377	\$120,689

Additionally, the Company has provided a guarantee on a loan for a joint venture. The Company's estimated maximum exposure under this arrangement at December 31, 2002 was approximately \$2 million.

During 2002, the Company purchased 501,330 shares of its common stock on the open market at a cost of \$11.6 million, bringing the total shares purchased to 7,740,789 at a cost of \$157.1 million through December 31, 2002. The Company is authorized to purchase up to an additional 2,259,211 shares under the share repurchase program. See Note B to the Consolidated Financial Statements.

A total of \$25.4 million in dividends was paid during 2002. In January 2003, the Company paid the quarterly cash dividend of 16 cents per share to shareholders of record on December 31, 2002.

Effective January 1, 2003, the Company will adopt the fair value method of recording stock options contained in SFAS No. 123, "*Accounting for Stock-Based Compensation*," which is considered the preferable accounting method for stock-based employee compensation. All future employee stock option grants beginning January 1, 2003 will be expensed over the stock option vesting period based on the fair value at the date the options are granted. The Company expects minimal financial impact in 2003 from the adoption of this accounting methodology. If the number of options granted in 2003 is comparable to 2002, the expected impact would be approximately \$0.01 per share for 2003. The Company estimates the impact per share would increase annually over the next several years and level off at approximately \$0.06 per share by 2006. This estimate assumes that the number and the fair value of options granted is similar for all years. The actual impact per share would be different in the event the number of options granted or the fair value of options increases or decreases from the current estimate. Historically, the Company has applied the intrinsic value method permitted under SFAS No. 123, as defined in Accounting Principles Board ("APB") Opinion No. 25, "*Accounting for Stock Issued to Employees*" and related interpretations, in accounting for the Company's stock option plans. Accordingly, no compensation cost has been recognized in the current or past years.

The Company accounts for its defined benefit plans in accordance with SFAS No. 87 "*Employers' Accounting for Pensions*". SFAS No. 87 requires a liability ("minimum pension liability") be recorded when the accumulated benefit obligation exceeds the fair value of plan assets. The Company has recently experienced a material decline in the fair market value of assets in the U.S. company's non-contributory defined benefit pension plan trust. This decline is due, in large part, to the generally weak economy and general declines in the market value of investments. At December 31, 2002, the Company recorded an after-tax charge to shareholders' equity of approximately \$80 million. In addition, the Company's annual pension expense for 2003 is expected to increase significantly. Based on December 31, 2002 pension investment values and expected actuarial assumptions used at December 31, 2002, the Company estimates pension expense will increase by \$15 million in 2003.

OFF-BALANCE SHEET FINANCIAL INSTRUMENTS

The Company utilizes letters of credit to back certain payment and performance obligations. Letters of credit are subject to limits based on amounts outstanding under the Company's Credit Agreement (see Note F). Outstanding letters of credit at December 31, 2002 were immaterial. The Company has also provided a guarantee on a loan for a joint venture. The Company believes the likelihood is remote that material payment will be required under this arrangement. The Company's estimated maximum exposure under this arrangement at December 31, 2002 was approximately \$2 million.

NEW ACCOUNTING PRONOUNCEMENTS

Effective January 1, 2002, the Company adopted SFAS No. 144, "*Accounting for the Impairment or Disposal of Long-Lived Assets*". SFAS No. 144 supercedes SFAS No. 121, "*Accounting for the Impairment of Long-Lived Assets and for Long-Lived Assets to be Disposed Of*" and APB Opinion No. 30, "*Reporting*

the Results of Operations — Reporting the Effects of Disposal of a Segment of a Business, and Extraordinary, Unusual and Infrequently Occurring Events”. SFAS No. 144 retains the requirements of SFAS No. 121 whereby an impairment loss is recognized if the carrying value of the asset is not recoverable from its undiscounted cash flows or fair values are less than carrying values. SFAS No. 144 broadens the scope of APB Opinion No. 30 provisions for the presentation of discontinued operations to include a component of an entity. The Statement requires a component of an entity sold or considered held for sale be presented as a discontinued operation. In addition, expected future operating losses from discontinued operations must be reflected in the periods incurred, rather than at the measurement date as previously required by APB Opinion No. 30. The adoption of this Statement did not have a material impact on the financial statements of the Company.

In April 2002, the FASB issued SFAS No. 145, “*Rescission of FASB Statements No. 4, 44 and 64, Amendment of FASB Statement No. 13, and Technical Corrections*”. SFAS No. 145 is effective for fiscal years beginning after May 15, 2002. SFAS No. 145 requires that debt extinguishment must meet the criteria under APB Opinion No. 30 to be classified as an extraordinary item. This statement also amends SFAS No. 13 to require sale-leaseback accounting for certain lease modifications that have economic effects similar to sale-leaseback transactions. The Company has not yet determined the impact, if any, this Statement will have on the financial statements of the Company.

In June 2002, the FASB issued SFAS No. 146, “*Accounting for Exit or Disposal Activities*”. SFAS No. 146 is effective for disposal activities initiated after December 31, 2002. SFAS No. 146 requires liabilities for one-time termination benefits incurred over future service periods be measured at fair value as of the termination date and recognized over the future service period. This Statement also requires liabilities associated with disposal activities be recorded when incurred instead of when probable as currently required by SFAS No. 5 “*Accounting for Contingencies*”. These liabilities should be adjusted for subsequent changes resulting from revisions to either the timing or amount of estimated cash flows, discounted at the original credit-adjusted risk-free rate. Interest on the liability would be accreted and charged to expense as an operating item. The Company has not yet determined the impact, if any, this Statement will have on the financial statements of the Company.

In November 2002, the FASB issued Interpretation No. 45, “*Guarantor’s Accounting and Disclosure Requirements for Guarantees, Including Indirect Guarantees of Indebtedness of Others*”. Interpretation No. 45 requires certain disclosures to be made by a guarantor in its financial statements about its obligations under certain guarantees, including product warranties. The disclosure provisions of this Statement are effective for the Company as of the fourth quarter of 2002. The Company has made the required disclosures in the financial statements. This Statement also requires a guarantor to recognize, at inception, for all guarantees issued or modified after December 31, 2002, a liability for the fair value of the obligations it has undertaken in issuing a guarantee. The adoption of the fair value provisions of this Statement are not expected to have a material impact on the financial statements of the Company.

In December 2002, the FASB issued SFAS No. 148, “*Accounting for Stock-Based Compensation — Transition and Disclosure*”. SFAS No. 148 does not alter the provisions of SFAS No. 123, nor does it require stock-based compensation to be measured under the fair-value method. Rather, SFAS No. 148 does provide three alternative transition methods to companies that elect to expense stock-based compensation using the fair-value approach under SFAS No. 123. As mentioned previously, the Company will adopt the fair-value method of recording stock options contained in SFAS No. 123 effective January 1, 2003, thus the provisions of this Statement will also be adopted by the Company at that date. The Company will elect to expense stock options using the prospective method prescribed in SFAS No. 148. The prospective method requires expense to be recognized for new grants or modifications issued beginning in the year of adoption. No expense is recognized in any year for options issued prior to adoption. The Company expects the impact from the adoption of this Statement to be approximately \$0.01 per share for 2003. The Company

estimates the impact per share would increase annually over the next several years and level off at approximately \$0.06 per share by 2006.

In December 2002, the Emerging Issues Task Force (“EITF”) issued EITF Issue 00-21, “ *Accounting for Revenue Arrangements with Multiple Deliverables*”. EITF 00-21 provides guidance on determining whether a revenue arrangement contains multiple deliverable items and if so, requires revenue be allocated amongst the different items based on fair value. EITF 00-21 also requires revenue on any item in a revenue arrangement with multiple deliverables not delivered completely must be deferred until delivery of the item is completed. The effective date of this Issue for the Company will be July 1, 2003. The Company has not yet determined the impact, if any, this Statement will have on the financial statements of the Company.

In January 2003, the FASB issued Interpretation No. 46, “ *Consolidation of Variable Interest Entities*”. Interpretation No. 46 provides guidance for identifying a controlling interest in a Variable Interest Entity (VIE) established by means other than voting interests. Interpretation No. 46 also requires consolidation of a VIE by an enterprise that holds such a controlling interest. The effective date for this Interpretation will be July 1, 2003. The Company has not yet determined the impact, if any, this Interpretation will have on the financial statements of the Company.

CRITICAL ACCOUNTING POLICIES

The Company’s consolidated financial statements are based on the selection and application of significant accounting policies, which require management to make estimates and assumptions. The Company believes the following are some of the more critical judgment areas in the application of its accounting policies that affect its financial condition and results of operations.

Legal And Tax Contingencies

The Company, like other manufacturers, is subject from time to time to a variety of civil and administrative proceedings arising in the ordinary course of business. Such claims and litigation include, without limitation, product liability claims and health, safety and environmental claims, some of which relate to cases alleging asbestos and manganese induced illnesses. The costs associated with these claims are predominantly defense costs, which are recognized in the periods incurred. Insurance reimbursements mitigate these costs and, where reimbursements are probable, they are recognized in the applicable period. With respect to costs other than defense costs (i.e. for liability and/or settlement or other resolution), reserves are recorded when it is probable that the contingencies will have an unfavorable outcome. The Company accrues its best estimate of the probable cost, after a review of the facts with management and counsel and taking into account past experience. If an unfavorable outcome is determined to be reasonably possible but not probable, or if the amount of loss cannot be reasonably estimated, disclosure is provided in lieu of reserves. Many of the current cases are in preliminary procedural stages and insufficient information exists upon which judgments can be made as to the validity or ultimate disposition of such actions. Therefore, a range of possible losses cannot be made at this time. Reserves are adjusted as facts and circumstances change, and related management assessments of the underlying merits and the likelihood of outcomes change. Moreover, reserves only cover identified and/or asserted claims. Future claims could, therefore, give rise to increases to such reserves. See Note M to the Consolidated Financial Statements and the Legal Proceedings section of the Annual Report on Form 10-K for further discussion of legal contingencies.

The Company often faces challenges from domestic and foreign tax authorities regarding the amount of taxes due. These challenges include questions regarding the timing and amount of deductions and the allocation of income among various tax jurisdictions. In evaluating the exposure associated with various filing positions, the Company records reserves for probable exposures. Based on evaluation of those tax positions, the Company believes it has appropriately accrued for probable exposures. To the extent the Company

were to prevail in matters for which accruals have been established or be required to pay amounts in excess of recorded reserves, the effective tax rate in a given financial statement period may be materially impacted.

Deferred Income Taxes

Deferred income taxes are recognized at currently enacted tax rates for temporary differences between the financial reporting and income tax bases of assets and liabilities and operating loss and tax credit carryforwards. The Company does not provide deferred income taxes on unremitted earnings of non-U.S. subsidiaries, as such funds are deemed permanently reinvested. As of December 31, 2002, the Company has approximately \$93 million of deferred tax assets related to deductible temporary differences and tax loss and credit carryforwards which will reduce taxable income in future years.

In assessing the realizability of deferred tax assets, the Company assesses whether it is more likely than not that a portion or all of the deferred tax assets will not be realized. The Company considers the scheduled reversal of deferred tax liabilities, projected future taxable income, and tax planning strategies in making this assessment. At December 31, 2002, a valuation allowance of \$15 million has been recorded against these deferred tax assets based on this assessment. The Company believes it is more likely than not that the tax benefit of the net deferred tax assets will be realized. The amount of net deferred tax asset considered realizable could be increased or reduced in the future if the Company’s assessment of future taxable income or tax planning strategies changes.

Pensions

The Company accounts for its defined benefit plans in accordance with SFAS No. 87, “*Employers’ Accounting for Pensions*” which requires amounts recognized in financial statements be determined on an actuarial basis. A substantial portion of the Company’s pension amounts relate to its defined benefit plan in the United States.

A significant element in determining the Company’s pension expense in accordance with SFAS No. 87 is the expected return on plan assets. The expected return on plan assets is determined based on the expected long-term rate of return on the plan assets and the market-related value of plan assets. The market-related value of plan assets can be determined by either fair value or a calculated value recognizing changes in fair value in a systematic and rational manner over not more than five years. The Company used fair values at December 31 for the market-related value of plan assets. The Company has assumed the expected long-term rate of return on plan assets will be 9.1%. The assumed long-term rate of return on assets is applied to the market value of plan assets. This produces the expected return on plan assets included in pension expense. The difference between this expected return and the actual return on plan assets is deferred. The amortization of the net deferral of past losses will increase future pension expense. The plan assets have earned a rate of return substantially less than 9.1% in the last two years. The Company anticipates lowering the expected rate of return on plan assets to approximately 8.8% for 2003. As a result, future pension expense will increase.

At the end of each year, the Company determines the discount rate to be used for plan liabilities. In estimating this rate, the Company looks to rates of return on high quality, fixed-income investments that receive one of the two highest ratings given by a recognized ratings agency. At December 31, 2002, the Company determined this rate to be 6.7%. Decreases in discount rates and lower expected returns on plan assets over the past two years will materially affect pension expense in 2003. The Company estimates pension expense will increase by \$15 million in 2003.

Inventories

Inventories are valued at the lower of cost or market. For domestic inventories, cost is determined principally by the last-in, first-out (LIFO) method, and for non-U.S. inventories, cost is determined by the first-in, first-out (FIFO) method. The valuation of LIFO inventories is made at the end of each year based on inventory levels and costs at that time. The excess of current cost over LIFO cost amounted to \$39.9 million at December 31, 2002. The Company reviews the net realizable value of inventory in detail on an on-going basis, with consideration given to deterioration, obsolescence and other factors. If actual market conditions differ from those projected by management, and the Company's estimates prove to be inaccurate, write-downs or adjustments to cost of sales may be required.

Accounts Receivable and Allowances

The Company maintains an allowance for doubtful accounts for estimated losses from the failure of its customers to make required payments for products delivered. The Company estimates this allowance based on knowledge of the financial condition of customers, review of historical receivables and reserve trends and other pertinent information. If the financial condition of customers deteriorates or an unfavorable trend in receivable collections is experienced in the future, additional allowances may be required.

CERTAIN FACTORS THAT MAY AFFECT FUTURE RESULTS

From time to time, information provided by the Company, statements by its employees or information included in its filings with the Securities and Exchange Commission (including those portions of this Management's Discussion and Analysis that refer to the future) may contain forward-looking statements that are not historical facts. Those statements are "forward-looking" within the meaning of the Private Securities Litigation Reform Act of 1995. All such forward-looking statements involve risks and uncertainties. Such forward-looking statements, and the Company's future performance, operating results, financial position and liquidity, are subject to a variety of factors that could materially affect future results, including:

- *Competition.* The Company operates in a highly competitive global environment and is subject to a variety of competitive factors such as pricing, the actions and strength of its competitors, and the Company's ability to maintain its position as a recognized leader in welding technology. The intensity of foreign competition is substantially affected by fluctuations in the value of the United States dollar against other currencies. The Company's competitive position could also be adversely affected should new or emerging entrants become more active in the arc welding business.
- *Economic and Market Conditions.* The Company is subject to general economic, business and industry conditions which can adversely affect the Company's results of operations. The Company's revenues and profits depend significantly on the overall demand for arc welding and cutting products. Capital spending in the manufacturing and other industrial sectors can adversely affect the Company's results of operations. If current economic and market conditions do not improve, the Company's results of operations could continue to be adversely affected.
- *International Markets.* The Company's long-term strategy is to increase its share in growing international markets, particularly Asia, Latin America, Central Europe and other developing markets. However, there can be no certainty that the Company will be successful in its expansion efforts. The Company is subject to the currency risks of doing business abroad, and the possible effects of international terrorism and hostilities. Moreover, international expansion poses challenging demands within the Company's infrastructure.

- *Cyclicality and Maturity of the Welding and Cutting Industry* . The United States arc welding and cutting industry is both mature and cyclical. The growth of the domestic arc welding and cutting industry has been and continues to be constrained by numerous factors, including increased cost of steel due to regulatory tariffs and the substitution of plastics and other materials in place of fabricated metal parts in many products and structures. Increased offshore production of fabricated steel structures has also decreased the domestic demand for arc welding and cutting products in the Company's largest market.
- *Litigation*. The Company, like other manufacturers, is subject in the U.S. market to a variety of product liability lawsuits and potential lawsuits that arise in the ordinary course of business. While past experience has generally shown these cases to be immaterial, product liability cases in the U.S. continue to increase generally, with unpredictable results.
- *Operating Factors*. The Company is highly dependent on its skilled workforce and efficient production facilities, which could be adversely affected by its labor relations, business interruptions and short-term or long-term interruptions in the availability of supplies or raw materials or in transportation of finished goods.
- *Research and Development* . The Company's continued success depends, in part, on its ability to continue to meet customer welding needs through the introduction of new products and the enhancement of existing product design and performance characteristics. There can be no assurances that new products or product improvements, once developed, will meet with customer acceptance and contribute positively to the operating results of the Company, or that product development will continue at a pace to sustain future growth.

The above list of factors that could materially affect the Company's future results is not all inclusive. Any forward-looking statements reflect only the beliefs of the Company or its management at the time the statement is made.

Item 7a. Qualitative and Quantitative Disclosure About Market Risk

The Company's primary financial market risks include fluctuations in currency exchange rates, commodity prices and interest rates. The Company manages these risks by using derivative financial instruments in accordance with established policies and procedures. The Company does not enter into derivatives or other financial instruments for trading or speculative purposes.

The Company enters into forward foreign exchange contracts principally to hedge the currency fluctuations in transactions denominated in foreign currencies, thereby limiting the Company's risk that would otherwise result from changes in exchange rates. During 2002, the principal transactions hedged were intercompany loans and intercompany purchases. The periods of the forward foreign exchange contracts correspond to the periods of the hedged transactions. Gains and losses on forward foreign exchange contracts and the offsetting losses and gains on hedged transactions are reflected in the income statement. At December 31, 2002, the Company had approximately \$26 million notional amount of foreign exchange contracts which primarily hedged recorded balance sheet exposures or intercompany purchases. The potential loss from a hypothetical 10% adverse change in foreign currency rates on the Company's open foreign exchange contracts at December 31, 2002 would not materially affect the Company's financial statements.

From time to time, the Company uses various hedging arrangements to manage the Company's exposure to price risk from commodity purchases. The primary commodities hedged are aluminum and copper. These hedging arrangements have the effect of locking in for specified periods (at predetermined prices or ranges of prices) the prices the Company will pay for the volume to which the hedge relates. The potential loss

from a hypothetical 10% adverse change in commodity prices on the Company's open commodity futures at December 31, 2002, would not materially affect the Company's financial statements.

The fair value of the Company's cash and cash equivalents at December 31, 2002, approximated carrying value due to its short-term duration. Market risk was estimated as the potential decrease in fair value resulting from a hypothetical 10% increase in interest rates for the issues contained in the investment portfolio and was not materially different from the year-end carrying value.

The Company uses floating rate swaps to convert a portion of its \$150 million fixed-rate, long-term borrowings into short-term variable interest rates. The weighted average interest rate on the Notes is 6.1% and the average maturity is eight years. At December 31, 2002, the Company's interest rate swaps had a notional amount of \$80 million with a fair value of \$9.9 million. The Company uses the short-cut method to account for these swaps as prescribed in SFAS No. 133, "Accounting for Derivative and Hedging Activities". The weighted average effective interest rate on the Notes after considering the effect of the interest rate swaps was 4.82% for 2002. A hypothetical decrease of 1% in the floating rate would not materially affect the Company's financial statements.

At December 31, 2002, the fair value of Notes payable to banks approximated carrying value due to its short-term maturities. Market risk was estimated as the potential increase in fair value resulting from a hypothetical 10% decrease in the Company's weighted-average short-term borrowing rate at December 31, 2002, and was not materially different from the year-end carrying value.

Item 8. Financial Statements and Supplementary Data

The response to this item is submitted in a separate section of this report following the signature page and certifications.

Item 9. Changes in and Disagreements with Accountants on Accounting and Financial Disclosure

None.

PART III

A definitive proxy statement will be filed pursuant to Regulation 14A of the Securities Exchange Act prior to April 30, 2003. Therefore, information required under this part, unless set forth below, is incorporated herein by reference from such definitive proxy statement.

Item 14. Controls and Procedures

Within the 90 days prior to the date of this Form 10-K, the Company carried out an evaluation, under the supervision and with the participation of the Company's management, including the Company's Chief Executive Officer and Chief Financial Officer, of the effectiveness of the design and operation of the Company's disclosure controls and procedures. Based on that evaluation, the Company's management, including the Chief Executive Officer and Chief Financial Officer, concluded that the Company's disclosure controls and procedures are operating effectively as designed. There have been no significant changes in the Company's internal controls or in other factors that could significantly affect internal controls subsequent to the date the Company carried out its evaluation.

PART IV

Item 15. Exhibits, Financial Statement Schedules and Reports on Form 8-K

(a)(1) Financial Statements

The following consolidated financial statements of the Company are included in a separate section of this report following the signature page and certifications:

- Report of Independent Auditors
- Consolidated Balance Sheets — December 31, 2002 and 2001
- Consolidated Statements of Income — Years ended December 31, 2002, 2001 and 2000
- Consolidated Statements of Shareholders’ Equity — Years ended December 31, 2002, 2001 and 2000
- Consolidated Statements of Cash Flows — Years ended December 31, 2002, 2001 and 2000
- Notes to Consolidated Financial Statements — December 31, 2002

(a)(2) Financial Statement Schedules

The following consolidated financial statement schedule of the Company is included in a separate section of this report following the signature page and certifications:

Schedule II — Valuation and Qualifying Accounts

All other schedules for which provision is made in the applicable accounting regulation of the Securities and Exchange Commission are not required under the related instructions or are inapplicable, and therefore, have been omitted.

(a)(3) Exhibits

Exhibit No.	Description
3(a)	Restated Articles of Incorporation of Lincoln Electric Holdings, Inc. (filed as Annex B to Form S-4 of Lincoln Electric Holdings, Inc., Registration No. 333-50435, filed on April 17, 1998, and incorporated herein by reference and made a part hereof).
3(b)	Amended Code of Regulations of Lincoln Electric Holdings, Inc. (filed as Exhibit 3(b) to Form 10-Q of Lincoln Electric Holdings, Inc. for the three months ended March 31, 2000, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(a)	Credit Agreement dated April 24, 2002 among Lincoln Electric Holdings, Inc., The Lincoln Electric Company, Lincoln Electric International Holding Company, Harris Calorific, Inc. and Lincoln Global, Inc. and the financial institutions listed in Annex A thereof, and KeyBank National Association, as Letter of Credit Issuer and Agent. (filed as Exhibit 10(r) to Form 10-Q of Lincoln Electric Holdings, Inc. for the three months ended March 31, 2002, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).

Exhibit No.	Description
10(b)	Note Purchase Agreement dated March 12, 2002 between Lincoln Electric Holdings, Inc. and The Lincoln Electric Company and the Purchasers listed in Schedule A thereof. (filed as Exhibit 10(q) to Form 10-Q of Lincoln Electric Holdings, Inc. for the three months ended March 31, 2002, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(c)	Amended and Restated Note Purchase and Private Shelf Agreement between Lincoln Electric Holdings, Inc., The Lincoln Electric Company and The Prudential Insurance Company of America dated as of April 30, 2002. (filed as Exhibit 10(v) to Form 10-Q of Lincoln Electric Holdings, Inc. for the three months ended June 30 2002, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(d)	Lincoln Electric Holdings, Inc. 1998 Stock Option Plan (filed as Annex F to the Registration Statement on Form S-4 of Lincoln Electric Holdings, Inc., Registration No. 333-50435, filed on April 17, 1998, incorporated herein by reference and made a part hereof).
10(e)	The Lincoln Electric Company 1988 Incentive Equity Plan (filed as Exhibit 28 to the Form S-8 Registration Statement of The Lincoln Electric Company, SEC File No. 33-25209 and incorporated herein by reference and made a part hereof) as adopted and amended by Lincoln Electric Holdings, Inc. pursuant to an Instrument of Adoption and Amendment dated December 29, 1998 (filed as Exhibit 10(d) to Form 10-K of Lincoln Electric Holdings, Inc. for the year ended December 31, 1998, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(f)	Form of Indemnification Agreement (filed as Exhibit A to The Lincoln Electric Company 1987 Proxy Statement, SEC File No. 0-1402, and incorporated herein by reference and made a part hereof).
10(g)	Amended and Restated Lincoln Electric Holdings, Inc. Supplemental Executive Retirement Plan, as amended (filed as Exhibit 10(u) to Form 10-Q of Lincoln Electric Holdings, Inc. for the three months ended March 31, 2002, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(h)	Amended and Restated Lincoln Electric Holdings, Inc. Deferred Compensation Plan dated January 1, 2002 (filed as Exhibit 10(s) to Form 10-Q of Lincoln Electric Holdings, Inc. for the three months ended March 31, 2002, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(i)	Description of Management Incentive Plan (filed as Exhibit 10(e) to Form 10-K of The Lincoln Electric Company for the year ended December 31, 1995, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(j)	Description of Long Term Performance Plan (filed as Exhibit 10(f) to Form 10-K of The Lincoln Electric Company for the year ended December 31, 1997, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).

Exhibit No.	Description
10(k)	Description of Non-Employee Directors' Restricted Stock Plan (filed as Exhibit 10(f) to Form 10-K of The Lincoln Electric Company for the year ended December 31, 1995, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof) as adopted by Lincoln Electric Holdings, Inc. pursuant to an Instrument of Adoption dated December 29, 1998 (filed as Exhibit 10(j) to Form 10-K of Lincoln Electric Holdings, Inc. for the year ended December 31, 1998, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(l)	The Lincoln Electric Company Non-Employee Directors' Deferred Compensation Plan (filed as Exhibit 10(g) to Form 10-K of The Lincoln Electric Company for the year ended December 31, 1995, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof) as amended by Amendment No. 1 dated as of December 29, 1998 (filed as Exhibit 10(k) to Form 10-K of Lincoln Electric Holdings, Inc. for the year ended December 31, 1998, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(m)	Letter of Employment between Anthony A. Massaro and Lincoln Electric Holdings, Inc. dated March 7, 2000 (filed as Exhibit 10(k) to Form 10-K of Lincoln Electric Holdings, Inc. for the year ended December 31, 1999, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(n)	Summary of Employment Agreements (filed as Exhibit 10(l) to Form 10-K of Lincoln Electric Holdings, Inc. for the year ended December 31, 1999, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(o)	The Lincoln Electric Company Executive Benefit Plan (filed as Exhibit 10(l) to Form 10-K of The Lincoln Electric Company for the year ended December 31, 1997, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof), as amend by Amendment No. 1 dated January 1, 2002 (filed as Exhibit 10(t) to Form 10-Q of Lincoln Electric Holdings, Inc. for the three months ended March 31, 2002, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(p)	Form of Severance Agreement (as entered into by the Company and the following executive officers: Mssrs. Massaro, Stropki, Elliott, Stueber and Vogt) (filed as Exhibit 10 to Form 10-Q of Lincoln Electric Holdings, Inc. for the nine months ended September 30, 1998, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(q)	Form of Amendment 1 to Severance Agreement (as entered into by the Company and the following executive officers: Messrs. Stropki and Stueber) (filed as Exhibit 10(o) to Form 10-K of Lincoln Electric Holdings, Inc. for the year ended December 31, 1999, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
10(r)	Stock Option Plan for Non-employee Directors (filed as Exhibit 10(p) to Form 10-Q of Lincoln Electric Holdings, Inc. for the three months ended March 31, 2000, SEC File No. 0-1402 and incorporated herein by reference and made a part hereof).
21	Subsidiaries of the Registrant.

Exhibit No.	Description
23	Consent of Independent Auditors.
24	Powers of Attorney.
99(1)	Certification pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002.
(b)	The Company did not file any reports on Form 8-K during the fourth quarter of 2002.
(c)	The exhibits which are listed under Item 15 (a) (3) are filed in a separate section of the report following the signature pages or incorporated by reference herein.
(d)	The financial statement schedule which is listed under item 15 (a) (2) is filed in a separate section of the report following the signature page and certifications.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

LINCOLN ELECTRIC HOLDINGS, INC.
(Registrant)

By: /s/ H. JAY ELLIOTT
H. Jay Elliott, Senior Vice President,
Chief Financial Officer and Treasurer
(principal financial and accounting officer)
February 14, 2003

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

/s/ ANTHONY A. MASSARO
Anthony A. Massaro, Chairman of the Board, President and Chief Executive Officer (principal executive officer)
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott, Senior Vice President, Chief Financial Officer and Treasurer (principal financial and accounting officer)
February 14, 2003

/s/ JOHN M. STROPKI
John M. Stropki, Director of the Company, Executive Vice President, President North America
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for HAROLD ADAMS
Harold Adams, Director
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for DAVID C. LINCOLN
David C. Lincoln, Director
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for HARRY CARLSON
Harry Carlson, Director
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for G. RUSSELL LINCOLN
G. Russell Lincoln, Director
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for RANKO CUCUZ
Ranko Cucuz, Director
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for KATHRYN JO LINCOLN
Kathryn Jo Lincoln, Director
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for DAVID H. GUNNING
David H. Gunning, Director
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for HENRY L. MEYER III
Henry L. Meyer III, Director
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for EDWARD E. HOOD, JR
Edward E. Hood, Jr., Director
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for HELLENE S. RUNTAGH
Hellene S. Runtagh, Director
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for PAUL E. LEGO
Paul E. Lego, Director
February 14, 2003

/s/ H. JAY ELLIOTT
H. Jay Elliott as Attorney-in-fact for FRANK L. STEINGASS
Frank L. Steingass, Director
February 14, 2003

Certification Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002

I, Anthony A. Massaro , Chairman, President and Chief Executive Officer of Lincoln Electric Holdings, Inc., certify that:

1. I have reviewed this annual report on Form 10-K of Lincoln Electric Holdings, Inc.;
2. Based on my knowledge, this annual report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this annual report;
3. Based on my knowledge, the financial statements, and other financial information included in this annual report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this annual report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-14 and 15d-14) for the registrant and we have:
 - a) designed such disclosure controls and procedures to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this annual report is being prepared;
 - b) evaluated the effectiveness of the registrant's disclosure controls and procedures as of a date within 90 days prior to the filing date of this annual report (the "Evaluation Date"); and
 - c) presented in this annual report our conclusions about the effectiveness of the disclosure controls and procedures based on our evaluation as of the Evaluation Date;
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation, to the registrant's auditors and the audit committee of registrant's board of directors (or persons performing the equivalent function):
 - a) all significant deficiencies in the design or operation of internal controls which could adversely affect the registrant's ability to record, process, summarize and report financial data and have identified for the registrant's auditors any material weaknesses in internal controls; and
 - b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal controls; and
6. The registrant's other certifying officer and I have indicated in this annual report whether or not there were significant changes in internal controls or in other factors that could significantly affect internal controls subsequent to the date of our most recent evaluation, including any corrective actions with regard to significant deficiencies and material weaknesses.

Date: February 14, 2003

/s/ Anthony A. Massaro

Anthony A. Massaro
Chairman, President and Chief Executive Officer

Certification Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002

I, H. Jay Elliott, Senior Vice President, Chief Financial Officer and Treasurer of Lincoln Electric Holdings, Inc., certify that:

1. I have reviewed this annual report on Form 10-K of Lincoln Electric Holdings, Inc.;
2. Based on my knowledge, this annual report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this annual report;
3. Based on my knowledge, the financial statements, and other financial information included in this annual report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this annual report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-14 and 15d-14) for the registrant and we have:
 - a) designed such disclosure controls and procedures to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this annual report is being prepared;
 - b) evaluated the effectiveness of the registrant's disclosure controls and procedures as of a date within 90 days prior to the filing date of this annual report (the "Evaluation Date"); and
 - c) presented in this annual report our conclusions about the effectiveness of the disclosure controls and procedures based on our evaluation as of the Evaluation Date;
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation, to the registrant's auditors and the audit committee of registrant's board of directors (or persons performing the equivalent function):
 - a) all significant deficiencies in the design or operation of internal controls which could adversely affect the registrant's ability to record, process, summarize and report financial data and have identified for the registrant's auditors any material weaknesses in internal controls; and
 - b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal controls; and
6. The registrant's other certifying officer and I have indicated in this annual report whether or not there were significant changes in internal controls or in other factors that could significantly affect internal controls subsequent to the date of our most recent evaluation, including any corrective actions with regard to significant deficiencies and material weaknesses.

Date: February 14, 2003

/s/ H. Jay Elliott

H. Jay Elliott
Senior Vice President, Chief Financial Officer and
Treasurer

ANNUAL REPORT ON FORM 10-K

ITEM 8, ITEM 15(a)(1) AND (2) AND ITEM 15(c) AND 15(d)

FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

FINANCIAL STATEMENT SCHEDULES

CERTAIN EXHIBITS

YEAR ENDED DECEMBER 31, 2002

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES

REPORT OF INDEPENDENT AUDITORS

Shareholders and Board of Directors
Lincoln Electric Holdings, Inc.

We have audited the accompanying consolidated balance sheets of Lincoln Electric Holdings, Inc. and subsidiaries as of December 31, 2002 and 2001, and the related consolidated statements of income, shareholders' equity and cash flows for each of the three years in the period ended December 31, 2002. Our audits also included the financial statement schedule listed in the Index at Item 15 (a)(2). These financial statements and schedule are the responsibility of the Company's management. Our responsibility is to express an opinion on these financial statements and schedule based on our audits.

We conducted our audits in accordance with auditing standards generally accepted in the United States. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement. An audit includes examining, on a test basis, evidence supporting the amounts and disclosures in the financial statements. An audit also includes assessing the accounting principles used and significant estimates made by management, as well as evaluating the overall financial statement presentation. We believe that our audits provide a reasonable basis for our opinion.

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the consolidated financial position of Lincoln Electric Holdings, Inc. and subsidiaries at December 31, 2002 and 2001, and the consolidated results of their operations and their cash flows for each of the three years in the period ended December 31, 2002, in conformity with accounting principles generally accepted in the United States. Also, in our opinion, the related financial statement schedule, when considered in relation to the basic financial statements taken as a whole, presents fairly in all material respects, the information set forth therein.

As discussed in Note A to the consolidated financial statements, effective January 1, 2002, the Company adopted Statement of Financial Accounting Standards No. 142, "Goodwill and Other Intangible Assets".

/s/ ERNST & YOUNG LLP

Cleveland, Ohio
January 29, 2003

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
CONSOLIDATED BALANCE SHEETS

	December 31	
	2002	2001
	(In thousands of dollars)	
ASSETS		
CURRENT ASSETS		
Cash and cash equivalents	\$176,076	\$ 23,493
Marketable securities	4,881	52
Accounts receivable (less allowances of \$6,805 in 2002; \$4,811 in 2001)	153,623	154,094
Inventories		
Raw materials and in-process	72,560	73,038
Finished goods	92,114	91,465
	164,674	164,503
Deferred income taxes	5,355	11,996
Other current assets	19,436	23,537
TOTAL CURRENT ASSETS	524,045	377,675
PROPERTY, PLANT AND EQUIPMENT		
Land	13,844	12,210
Buildings	147,631	134,981
Machinery and equipment	457,060	435,074
	618,535	582,265
Less: accumulated depreciation and amortization	346,682	311,294
	271,853	270,971
OTHER ASSETS		
Prepaid pension costs	3,345	38,784
Equity investments in affiliates	29,735	26,157
Intangibles, net	12,754	5,846
Goodwill	4,095	40,416
Deferred income taxes	17,858	—
Other	37,584	21,462
	105,371	132,665
TOTAL ASSETS	\$901,269	\$781,311

See notes to these consolidated financial statements

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
CONSOLIDATED BALANCE SHEETS

	December 31	
	2002	2001
	(In thousands of dollars, except share data)	
LIABILITIES AND SHAREHOLDERS' EQUITY		
CURRENT LIABILITIES		
Notes payable to banks	\$ 945	\$ 13,163
Trade accounts payable	63,894	67,569
Accrued employee compensation and benefits	38,663	26,456
Accrued expenses	12,722	13,519
Taxes, including income taxes	34,699	34,216
Dividend payable	6,746	6,355
Other current liabilities	20,514	22,788
Current portion of long-term debt	13,433	13,434
TOTAL CURRENT LIABILITIES	191,616	197,500
Long-term debt, less current portion	174,146	24,181
Accrued pensions	92,066	17,849
Deferred income taxes	1,070	31,751
Other long-term liabilities	13,218	11,332
SHAREHOLDERS' EQUITY		
Preferred Shares, without par value — at stated capital amount:		
Authorized — 5,000,000 shares in 2002 and 2001; Issued and Outstanding — none	—	—
Common Shares, without par value — at stated capital amount:		
Authorized — 120,000,000 shares in 2002 and 2001; Issued — 49,282,306 shares in 2002 and 2001; Outstanding — 42,087,115 shares in 2002 and 42,369,145 shares in 2001	4,928	4,928
Additional paid-in capital	106,237	105,380
Retained earnings	597,495	594,701
Accumulated other comprehensive income (loss)	(132,350)	(66,726)
Treasury shares, at cost — 7,195,191 shares in 2002 and 6,913,161 shares in 2001	(147,157)	(139,585)
TOTAL SHAREHOLDERS' EQUITY	429,153	498,698
TOTAL LIABILITIES AND SHAREHOLDERS' EQUITY	\$ 901,269	\$ 781,311

See notes to these consolidated financial statements.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF INCOME

	Year Ended December 31		
	2002	2001	2000
	(In thousands of dollars, except per share data)		
Net sales	\$994,077	\$978,877	\$1,058,601
Cost of goods sold	694,052	671,551	703,503
Gross profit	300,025	307,326	355,098
Selling, general & administrative expenses	198,035	190,161	216,217
Rationalization charges	10,468	—	—
Operating income	91,522	117,165	138,881
Other income (expense):			
Interest income	3,239	1,122	732
Other income (expense)	2,238	3,291	(10,553)
Interest expense	(9,056)	(5,537)	(7,383)
Total other income (expense)	(3,579)	(1,124)	(17,204)
Income before income taxes and the cumulative effect of a change in accounting principle	87,943	116,041	121,677
Income taxes	21,061	32,452	43,585
Income before the cumulative effect of a change in accounting principle	66,882	83,589	78,092
Cumulative effect of a change in accounting principle, net of tax	(37,607)	—	—
Net income	\$ 29,275	\$ 83,589	\$ 78,092
Per share amounts:			
Basic earnings per share			
Basic earnings per share before the cumulative effect of a change in accounting principle	\$ 1.58	\$ 1.97	\$ 1.83
Cumulative effect of a change in accounting principle, net of tax	(0.89)	—	—
Basic earnings per share	\$ 0.69	\$ 1.97	\$ 1.83
Diluted earnings per share			
Diluted earnings per share before the cumulative effect of a change in accounting principle	\$ 1.56	\$ 1.96	\$ 1.83
Cumulative effect of a change in accounting principle, net of tax	(0.88)	—	—
Diluted earnings per share	\$ 0.68	\$ 1.96	\$ 1.83

See notes to these consolidated financial statements.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF SHAREHOLDERS' EQUITY

(in thousands of dollars, except per share data)	Comm Shares	Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income (Loss)	Treasury Shares	Total
Balance January 1, 2000	\$4,928	\$104,891	\$483,463	\$ (43,524)	\$ (98,243)	\$451,515
Comprehensive income						
Net income			78,092			78,092
Minimum pension liability adjustment				(429)		(429)
Currency translation adjustment				(16,035)		(16,035)
Total comprehensive income						61,628
Cash dividends declared — \$0.57 per share			(24,157)			(24,157)
Net shares issued under certain benefit plans		2	(127)		645	520
Purchase of shares for treasury					(42,205)	(42,205)
Balance December 31, 2000	4,928	104,893	537,271	(59,988)	(139,803)	447,301
Comprehensive income						
Net income			83,589			83,589
Minimum pension liability adjustment				(514)		(514)
Unrealized gain on derivatives designated and qualified as cash flow hedges, net of tax				226		226
Currency translation adjustment				(6,450)		(6,450)
Total comprehensive income						76,851
Cash dividends declared — \$0.60 per share			(25,422)			(25,422)
Net shares issued under certain benefit plans		487	(737)		2,643	2,393
Purchase of shares for treasury					(2,425)	(2,425)
Balance December 31, 2001	4,928	105,380	594,701	(66,726)	(139,585)	498,698
Comprehensive income						
Net income			29,275			29,275
Minimum pension liability adjustment, net of tax of \$48,206				(79,697)		(79,697)
Unrealized loss on derivatives designated and qualified as cash flow hedges, net of tax				(246)		(246)
Currency translation adjustment				14,319		14,319
Total comprehensive income (loss)						(36,349)
Cash dividends declared — \$0.61 per share			(25,769)			(25,769)
Net shares issued under certain benefit plans		857	(712)		4,018	4,163
Purchase of shares for treasury					(11,590)	(11,590)
Balance December 31, 2002	\$4,928	\$106,237	\$597,495	\$(132,350)	\$(147,157)	\$429,153

See notes to these consolidated financial statements.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF CASH FLOWS

	Year Ended December 31		
	2002	2001	2000
	(In thousands of dollars)		
OPERATING ACTIVITIES			
Net income	\$ 29,275	\$ 83,589	\$ 78,092
Adjustments to reconcile net income to net cash provided by operating activities:			
Cumulative effect of a change in accounting principle, net of tax	37,607	—	—
Rationalization charges	10,468	—	—
Depreciation and amortization	37,040	36,205	34,712
Deferred income taxes	9,021	16,527	(2,286)
Loss (gain) on sale of fixed assets	476	(2,574)	(520)
Changes in operating assets and liabilities net of effects from acquisitions:			
Decrease (increase) in accounts receivable	8,786	(742)	1,905
Decrease in inventories	6,636	18,653	6,005
Decrease (increase) in other current assets	5,452	2,381	(4,659)
(Decrease) increase in accounts payable	(8,126)	4,647	(4,963)
(Decrease) increase in other current liabilities	(2,032)	(25,495)	12,030
(Decrease) in prepaid pension expense	(21,400)	(12,079)	(7,223)
Gross change in other long-term assets and liabilities	(2,005)	(775)	(305)
Other, net	(7,600)	(136)	8,046
NET CASH PROVIDED BY OPERATING ACTIVITIES	103,598	120,201	120,834
INVESTING ACTIVITIES			
Capital expenditures	(27,909)	(36,723)	(34,800)
Acquisitions of businesses and equity investments	(8,010)	(3,961)	(18,903)
Proceeds from sale of fixed assets	2,052	4,420	1,627
Purchase of marketable securities	(5,015)	(10)	—
Other	391	316	6
NET CASH (USED) BY INVESTING ACTIVITIES	(38,491)	(35,958)	(52,070)
FINANCING ACTIVITIES			
Proceeds from short-term borrowings	3,688	50,542	47,046
Payments on short-term borrowings	(10,926)	(44,299)	(48,972)
Notes payable to banks — net	(7,114)	(35,280)	29,270
Proceeds from long-term borrowings	150,172	71	54,294
Payments on long-term borrowings	(13,661)	(17,410)	(80,266)
Issuance of shares from treasury	3,294	1,891	645
Purchase of shares for treasury	(11,590)	(2,427)	(42,205)
Cash dividends paid	(25,390)	(25,418)	(24,034)
Other	—	(15)	(442)
NET CASH PROVIDED (USED) BY FINANCING ACTIVITIES	88,473	(72,345)	(64,664)
Effect of exchange rate changes on cash and cash equivalents	(997)	276	(1,456)
INCREASE IN CASH AND CASH EQUIVALENTS	152,583	12,174	2,644
Cash and cash equivalents at beginning of year	23,493	11,319	8,675
CASH AND CASH EQUIVALENTS AT END OF YEAR	\$176,076	\$ 23,493	\$ 11,319

See notes to these consolidated financial statements.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS
(In thousands of dollars except share and per share data)
December 31, 2002

NOTE A — SIGNIFICANT ACCOUNTING POLICIES

Principles of Consolidation: The consolidated financial statements include the accounts of Lincoln Electric Holdings, Inc., its wholly-owned and majority-owned subsidiaries and all non-majority owned entities for which it has a controlling interest (the “Company”) after elimination of all significant intercompany accounts, transactions and profits. Minority ownership interest in consolidated subsidiaries, which is not material, is recorded in Other long-term liabilities.

Cash Equivalents: The Company considers all highly liquid investments with a maturity of three months or less when purchased to be cash equivalents.

Inventories: Inventories are valued at the lower of cost or market. For domestic inventories, cost is determined principally by the last-in, first-out (LIFO) method, and for non-U.S. inventories, cost is determined by the first-in, first-out (FIFO) method. At December 31, 2002 and 2001, approximately 53% and 57%, respectively, of total inventories were valued using the LIFO method. The excess of current cost over LIFO cost amounted to \$39,924 at December 31, 2002 and \$39,703 at December 31, 2001.

Equity Investments: Investments in businesses in which the Company does not have a controlling interest and holds between a 20% and 50% ownership interest are accounted for using the equity method of accounting. Under the equity method, the investment is carried at cost plus the Company’s proportionate share of the net income or loss of the business since the date of acquisition.

Property, Plant and Equipment: Property, plant and equipment are stated at cost and include improvements which significantly extend the useful lives of existing plant and equipment. Depreciation and amortization are computed by both accelerated and straight-line methods over useful lives ranging from 3 to 20 years for machinery, tools and equipment, and up to 50 years for buildings. Net gains or losses related to asset dispositions are recognized in earnings in the period in which dispositions occur.

Goodwill and Intangibles: Prior to January 1, 2002, the Company amortized goodwill on a straight line basis over periods not exceeding 40 years. Goodwill had previously been tested for impairment under the provisions of Statement of Financial Accounting Standards (SFAS) No. 121, “*Accounting for the Impairment of Long-lived Assets and Long-lived Assets to be Disposed of*”. The Company previously evaluated goodwill for impairment by comparing the unamortized balance of goodwill to projected undiscounted cash flows, which did not indicate an impairment. Effective January 1, 2002, the Company adopted SFAS No. 142, “ *Goodwill and Other Intangible Assets*”. SFAS No. 142 requires cessation of goodwill amortization and a fair value approach to testing the impairment of goodwill and other intangibles. The Company has performed the required impairment tests of goodwill and indefinite-lived intangible assets as of January 1, 2002. As a result of these impairment tests, including third-party valuations based on acquisition prices of comparable businesses, a non-cash goodwill impairment charge of \$37,607 or \$0.88 per diluted share, net of tax benefits of \$700, has been recorded as a cumulative effect of a change in accounting principle. This non-cash charge was primarily due to weak market conditions in the European region and the Company’s expectations of lower financial performance at these reporting units. The Company performed its annual impairment test on October 1, 2002 and determined there was no additional impairment of the remaining goodwill.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

The following table reflects net income adjusted to exclude amortization of goodwill and the cumulative effect of a change in accounting principle, net of tax, in the periods presented:

(\$000's, except for earnings per share amounts)	For the Year Ended December 31,		
	2002	2001	2000
Net income, as reported	\$29,275	\$83,589	\$78,092
Add: Goodwill amortization	—	1,648	1,623
Adjusted net income	29,275	85,237	79,715
Add: Cumulative effect of a change in accounting principle, net of tax	37,607	—	—
Adjusted income before the cumulative effect of a change in accounting principle	\$66,882	\$85,237	\$79,715

The following table reflects Earnings Per Share adjusted to exclude amortization of goodwill and the cumulative effect of a change in accounting principle, net of tax, in the periods presented:

	For the Year Ended December 31,		
	2002	2001	2000
<u>Basic Earnings Per Share:</u>			
Net income, as reported	\$0.69	\$1.97	\$1.83
Add: Goodwill amortization	—	0.04	0.04
Adjusted net income	0.69	2.01	1.87
Add: Cumulative effect of a change in accounting principle, net of tax	0.89	—	—
Adjusted income before the cumulative effect of a change in accounting principle	\$1.58	\$2.01	\$1.87
<u>Diluted Earnings Per Share:</u>			
Net income, as reported	\$0.68	\$1.96	\$1.83
Add: Goodwill amortization	—	0.04	0.04
Adjusted net income	0.68	2.00	1.87
Add: Cumulative effect of a change in accounting principle, net of tax	0.88	—	—
Adjusted income before the cumulative effect of a change in accounting principle	\$1.56	\$2.00	\$1.87

The changes in the carrying amount of net goodwill by segment for the year ended December 31, 2002 are as follows:

	United States	Europe	Other Countries	Consolidated Total
Balance as of January 1, 2002	\$ 22,688	\$ 17,027	\$ 701	\$ 40,416
Foreign exchange effect on prior balances		(102)	7	(95)
Goodwill acquired during the period	—	1,982	99	2,081
Impairment loss from adoption of SFAS No. 142	(22,688)	(14,812)	(807)	(38,307)
Balance as of December 31, 2002	\$ —	\$ 4,095	\$ —	\$ 4,095

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

Gross intangible assets as of December 31, 2002 and 2001 were \$21,992 and \$12,907, respectively, which included accumulated amortization of \$9,238 and \$7,060, respectively. Aggregate amortization expense was \$1,028, \$1,066 and \$960 for 2002, 2001 and 2000, respectively.

Estimated annual intangible amortization expense in each of the next five years ending December 31 will be approximately \$1,000.

Long-lived Assets: During the first quarter of 2002, the Company adopted SFAS No. 144, “ *Accounting for the Impairment or Disposal of Long-Lived Assets*”. The Company previously evaluated long-lived assets for impairment under SFAS No. 121. SFAS No. 144 retains the requirements of SFAS No. 121 whereby the carrying value of long-lived assets is reviewed if facts and circumstances indicate a potential impairment of carrying value may have occurred utilizing relevant cash flow and profitability information. Impairment losses are recorded when the undiscounted cash flows estimated to be generated by those assets are less than the assets’ carrying amounts. The adoption of this Statement did not have a material impact on the financial statements of the Company.

Product Warranties: The Company accrues for product warranty claims based on historical experience and the expected material and labor costs to provide warranty service. The changes in the carrying amount of Product Warranty reserves for the year ended December 31, 2002 is as follows:

Balance at beginning of period	Charged to costs and expenses	Deductions	Balance at end of period
\$6,254	\$3,121	\$(3,363)	\$6,012

Revenue Recognition: The Company recognizes revenue when product is shipped and title is transferred to the customer.

Distribution Costs: Distribution costs, including warehousing and freight related to product shipments, is included in Cost of goods sold.

Stock-Based Compensation: Currently, the Company applies the intrinsic value method permitted under SFAS No. 123, “ *Accounting for Stock-Based Compensation*,” as defined in Accounting Principles Board (“APB”) Opinion No. 25, “ *Accounting for Stock Issued to Employees*” and related interpretations, in accounting for the Company’s stock option plans. Accordingly, no compensation cost has been recognized in the current or past years.

SFAS No. 123 requires pro forma disclosure of the effect on net income and earnings per share when applying the fair value method of valuing stock-based compensation. The following table sets forth the pro forma disclosure of net income and earnings per share using the Black-Scholes option pricing model. For purposes of this pro forma disclosure, the estimated fair value of the options is amortized ratably over the vesting periods.

	2002		2001		2000	
	Pro Forma	Reported	Pro Forma	Reported	Pro Forma	Reported
Net income	\$26,254	\$29,275	\$81,167	\$83,589	\$76,413	\$78,092
Basic earnings per share	0.62	0.69	1.92	1.97	1.79	1.83
Diluted earnings per share	0.61	0.68	1.90	1.96	1.79	1.83

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

Effective January 1, 2003, the Company will adopt the fair value method of recording stock options contained in SFAS No. 123, which is considered the preferable accounting method for stock-based employee compensation. All future employee stock option grants beginning January 1, 2003 will be expensed over the stock option vesting period based on the fair value at the date the options are granted. The Company will elect to expense stock options using the prospective method prescribed in SFAS No. 148, “ *Accounting for Stock-Based Compensation — Transition and Disclosure*”. The prospective method requires expense to be recognized for new grants or modifications issued beginning in the year of adoption. No expense is recognized in any year for options issued prior to adoption. The Company expects the impact from the adoption of this Statement to be approximately \$0.01 per share for 2003. The Company estimates the impact per share would increase annually over the next several years and level off at approximately \$0.06 per share by 2006.

Translation of Foreign Currencies: Asset and liability accounts are translated into U.S. dollars using exchange rates in effect at the date of the consolidated balance sheet; revenue and expense accounts are translated at monthly exchange rates. Translation adjustments are reflected as a component of Shareholders’ Equity. For subsidiaries operating in highly inflationary economies, both historical and current exchange rates are used in translating balance sheet accounts, and translation adjustments are included in net income.

Transaction gains and losses are included in Selling, general & administrative expenses and were not material.

Financial Instruments: The Company, on a limited basis, uses forward exchange contracts to hedge exposure to exchange rate fluctuations on certain intercompany loans, purchase and sales transactions and other intercompany commitments. Contracts are written on a short-term basis and are not held for trading or speculative purposes.

The Company recognizes derivative instruments as either assets or liabilities in the balance sheets at fair value. The accounting for changes in the fair value of derivative instruments depends on whether it has been designated and qualifies as part of a hedging relationship and further, on the type of hedging relationship.

For derivative instruments that qualify as a fair value hedge (i.e., hedging the exposure to changes in the fair value of an asset or a liability), the gain or loss on the derivative instrument, as well as the offsetting loss or gain on the hedged item attributable to the hedged risk are recognized in earnings. For derivative instruments that qualify as a cash flow hedge (i.e., hedging the exposure to variability in expected future cash flows), the effective portion of the gain or loss on the derivative instrument is reported as a component of Other comprehensive income and reclassified into earnings in the same period or periods during which the hedged transaction affects earnings. Any remaining gain or loss on the derivative instrument is recognized in earnings. The Company does not hedge its net investments in foreign subsidiaries. For derivative instruments not designated as hedges, the gain or loss from changes in their fair values is recognized in earnings.

Research and Development: Research and development costs are expensed as incurred, and totaled \$19,150 in 2002, \$17,946 in 2001 and \$16,604 in 2000.

Estimates: The preparation of financial statements in conformity with accounting principles generally accepted in the United States, requires management to make estimates and assumptions in certain circumstances that affect the amounts reported in the accompanying consolidated financial statements and notes. Actual results could differ from these estimates.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

Earnings per Share: The following table sets forth the computation of basic and diluted earnings per share (dollars and shares in thousands, except per share amounts).

	2002	2001	2000
Numerator:			
Income before the cumulative effect of a change in accounting principle	\$ 66,882	\$83,589	\$78,092
Cumulative effect of a change in accounting principle, net of tax	(37,607)	—	—
Net income	\$ 29,275	\$83,589	\$78,092
Denominator:			
Denominator for basic earnings per share - Weighted-average shares outstanding	42,259	42,375	42,670
Effect of dilutive securities — Employee stock options	540	235	20
Denominator for diluted earnings per share - Adjusted weighted-average shares outstanding	42,799	42,610	42,690
Basic earnings per share			
Income before the cumulative effect of a change in accounting principle	\$ 1.58	\$ 1.97	\$ 1.83
Cumulative effect of a change in accounting principle, net of tax	(0.89)	—	—
Basic earnings per share	\$ 0.69	\$ 1.97	\$ 1.83
Diluted earnings per share			
Income before the cumulative effect of a change in accounting principle	\$ 1.56	\$ 1.96	\$ 1.83
Cumulative effect of a change in accounting principle, net of tax	(0.88)	—	—
Diluted earnings per share	\$ 0.68	\$ 1.96	\$ 1.83

Reclassification: Certain reclassifications have been made to prior-year financial statements to conform to current year classifications.

New Accounting Pronouncements: Effective January 1, 2002, the Company adopted SFAS No. 144, “ *Accounting for the Impairment or Disposal of Long-Lived Assets*”. SFAS No. 144 supercedes SFAS No. 121, and APB Opinion No. 30, “ *Reporting the Results of Operations — Reporting the Effects of Disposal of a Segment of a Business, and Extraordinary, Unusual and Infrequently Occurring Events*”. SFAS No. 144 retains the requirements of SFAS No. 121 whereby an impairment loss is recognized if the carrying value of the asset is not recoverable from its undiscounted cash flows or fair values are less than carrying values. SFAS No. 144 broadens the scope of APB Opinion No. 30 provisions for the presentation of discontinued operations to include a component of an entity. The Statement requires a component of an entity sold or considered held for sale be presented as a discontinued operation. In addition, expected future operating losses from discontinued operations must be reflected in the periods incurred, rather than at the measurement date as previously required by APB Opinion No. 30. The adoption of this Statement did not have a material impact on the financial statements of the Company.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

In April 2002, the FASB issued SFAS No. 145, “*Rescission of FASB Statements No. 4, 44 and 64, Amendment of FASB Statement No. 13, and Technical Corrections*”. SFAS No. 145 is effective for fiscal years beginning after May 15, 2002. SFAS No. 145 requires that debt extinguishment must meet the criteria under APB Opinion No. 30 to be classified as an extraordinary item. This statement also amends SFAS No. 13 to require sale-leaseback accounting for certain lease modifications that have economic effects similar to sale-leaseback transactions. The Company has not yet determined the impact, if any, this Statement will have on the financial statements of the Company.

In June 2002, the FASB issued SFAS No. 146, “*Accounting for Exit or Disposal Activities*”. SFAS No. 146 is effective for disposal activities initiated after December 31, 2002. SFAS No. 146 requires liabilities for one-time termination benefits incurred over future service periods be measured at fair value as of the termination date and recognized over the future service period. This Statement also requires liabilities associated with disposal activities be recorded when incurred instead of when probable as currently required by SFAS No. 5 “*Accounting for Contingencies*”. These liabilities should be adjusted for subsequent changes resulting from revisions to either the timing or amount of estimated cash flows, discounted at the original credit-adjusted risk-free rate. Interest on the liability would be accreted and charged to expense as an operating item. The Company has not yet determined the impact, if any, this Statement will have on the financial statements of the Company.

In November 2002, the FASB issued Interpretation No. 45, “*Guarantor’s Accounting and Disclosure Requirements for Guarantees, Including Indirect Guarantees of Indebtedness of Others*”. Interpretation No. 45 requires certain disclosures to be made by a guarantor in its financial statements about its obligations under certain guarantees, including product warranties. The disclosure provisions of this Statement are effective for the Company as of the fourth quarter of 2002. The Company has made the required disclosures in these financial statements. This Statement also requires a guarantor to recognize, at inception, for all guarantees issued or modified after December 31, 2002, a liability for the fair value of the obligations it has undertaken in issuing a guarantee. The adoption of the fair value provisions of this Statement are not expected to have a material impact on the financial statements of the Company.

In December 2002, the FASB issued SFAS No. 148, “*Accounting for Stock-Based Compensation — Transition and Disclosure*”. SFAS No. 148 does not alter the provision of SFAS No. 123, nor does it require stock-based compensation to be measured under the fair-value method. Rather, SFAS No. 148 does provide three alternative transition methods to companies that elect to expense stock-based compensation using the fair-value approach under SFAS No. 123. As mentioned previously, the Company will adopt the fair-value method of recording stock options contained in SFAS No. 123 effective January 1, 2003. The Company will elect to expense stock options using the prospective method prescribed in SFAS No. 148. The prospective method requires expense to be recognized for new grants or modifications issued beginning in the year of adoption. No expense is recognized in any year for options issued prior to adoption. The Company expects the impact from the adoption of this Statement to be approximately \$0.01 per share for 2003. The Company estimates the impact per share would increase annually over the next several years and level off at approximately \$0.06 per share by 2006.

In December 2002, the Emerging Issues Task Force (“EITF”) issued EITF Issue 00-21, “*Accounting for Revenue Arrangements with Multiple Deliverables*”. EITF 00-21 provides guidance on determining whether a revenue arrangement contains multiple deliverable items and if so, requires revenue be allocated amongst the different items based on fair value. EITF 00-21 also requires revenue on any item in a revenue arrangement with multiple deliverables not delivered completely must be deferred until delivery of the item is completed. The effective date of this Issue for the Company will be July 1, 2003. The Company has not yet determined the impact, if any, this Statement will have on the financial statements of the Company.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

In January 2003, the FASB issued Interpretation No. 46, “ *Consolidation of Variable Interest Entities*”. Interpretation No. 46 provides guidance for identifying a controlling interest in a Variable Interest Entity (“VIE”) established by means other than voting interests. Interpretation No. 46 also requires consolidation of a VIE by an enterprise that holds such a controlling interest. The effective date for this Interpretation will be July 1, 2003. The Company has not yet determined the impact, if any, this Interpretation will have on the financial statements of the Company.

Other: Included in Selling, general & administrative expenses are the costs related to the Company’s discretionary employee bonus, net of hospitalization costs, of \$32,218 in 2002, \$31,947 in 2001 and \$54,509 in 2000.

NOTE B — SHAREHOLDERS’ EQUITY

In 1999, the Board of Directors authorized an additional share repurchase program of up to 5,000,000 shares of the Company’s outstanding Common Shares to satisfy obligations under its stock option plans. This share repurchase program is in addition to the 5,000,000 shares authorized for repurchase by the Board of Directors in September 1998. The Company purchased 501,330 shares at an average cost of \$23.12 per share in 2002, bringing the total shares purchased through December 31, 2002, to 7,740,789 at an average cost of \$20.30 per share.

NOTE C — ACCUMULATED OTHER COMPREHENSIVE INCOME (LOSS)

The components of accumulated other comprehensive income (loss) are as follows:

	Minimum Pension Liability Adjustment, net of tax	Currency Translation Adjustment	Unrealized Gain (Loss) on Derivatives Designated and Qualified as Cash Flow Hedges, net of tax	Total Accumulated Other Comprehensive Loss
Balance January 1, 2000	\$ (792)	\$(42,732)	\$ —	\$ (43,524)
Other comprehensive (loss)	(429)	(16,035)	—	(16,464)
Balance December 31, 2000	(1,221)	(58,767)	—	(59,988)
Other comprehensive (loss) income	(514)	(6,450)	226	(6,738)
Balance December 31, 2001	(1,735)	(65,217)	226	(66,726)
Other comprehensive (loss) income	(79,697)	14,319	(246)	(65,624)
Balance December 31, 2002	\$(81,432)	\$(50,898)	\$ (20)	\$(132,350)

NOTE D — STOCK PLANS

The 1998 Stock Option Plan (“Stock Option Plan”) was adopted by the shareholders to replace The Lincoln Electric Company 1988 Incentive Equity Plan (“Incentive Equity Plan”) which expired in May 1998. The Stock Option Plan provides for the grant of options for 5,000,000 shares of Company stock to key employees over a ten-year period.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE D — STOCK PLANS (Continued)

The following table summarizes the option activity for the three years ended December 31, 2002, under all Plans:

	2002		2001		2000	
	Options	Weighted-Average Exercise Price	Options	Weighted-Average Exercise Price	Options	Weighted-Average Exercise Price
Balance, beginning of year	2,736,251	\$18.12	2,218,793	\$16.76	1,567,334	\$18.13
Options granted	669,900	\$22.97	694,390	\$21.65	787,375	\$13.60
Options exercised	(222,246)	\$15.21	(171,082)	\$14.91	(29,876)	\$14.08
Options canceled	(4,434)	\$19.57	(5,850)	\$17.56	(106,040)	\$14.36
Balance, end of year	3,179,471	\$19.34	2,736,251	\$18.12	2,218,793	\$16.76
Exercisable at end of year	1,875,464	\$18.29	1,430,280	\$17.75	1,065,512	\$17.82

During 1996, options for 335,180 shares were granted to employees in settlement of a lawsuit over performance awards relating to prior years. Exercise prices are \$15.00 and \$17.00 per share. These options are exercisable over five- and ten-year periods and are fully vested, non-qualified and non-transferable. At December 31, 2002 and 2001, there were 94,756 and 124,554, respectively, of these options outstanding.

All other options granted under both the Stock Option Plan and the Incentive Equity Plan are outstanding for a term of ten years from the date of grant. The majority of the options granted under both plans vest ratably over a period of three years from the grant date. The exercise prices of all options were equal to the fair market value of the Company’s shares at the date of grant. As permitted under SFAS No. 123, the Company has continued to record stock-based compensation in accordance with the intrinsic value method established by APB Opinion No. 25. Under the intrinsic value method, compensation expense is measured as the excess, if any, of the market price at the date of grant over the exercise price of the options. Accordingly, no compensation expense was recognized upon the award of these stock options.

In estimating the fair value of options granted during 2002 for the Stock Option Plan and the Incentive Equity Plan, an expected option life of five years was used based on the Company’s historical experience. In prior years, the Company had used ten years as the expected option life. The other weighted-average assumptions were as follows:

	2002	2001	2000
Expected volatility	43.50%	43.13%	42.60%
Dividend yield	2.76%	2.45%	2.90%
Risk-free interest rate	3.60%	5.11%	5.17%
Weighted-average fair value of options granted during the year	\$ 7.63	\$ 9.52	\$ 5.59

The Stock Option Plan for Non-Employee Directors (“Directors Stock Option Plan”) was adopted in May 2000 to replace The Lincoln Non-Employee Directors Restricted Stock Plan, which was terminated. The Directors Stock Option Plan provides for the grant of stock options for the purchase of up to an aggregate of 500,000 Common Shares. Options issued under the Directors Stock Option Plan were 30,000 in 2002 and 34,000 in 2001. There were no shares issued in connection with The Lincoln Non-Employee Directors Restricted Stock Plan during 2002 or 2001 and 5,335 in 2000. Forfeitures under the service requirements of the plan were 1,644 in 2000.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE D — STOCK PLANS (Continued)

At December 31, 2002, there were 2,594,535 shares of common stock available for future grant under all plans, and the weighted-average remaining contractual life of outstanding options was 7.6 years. The following table summarizes information about stock options outstanding as of December 31, 2002:

Exercise Price Range	Outstanding		Exercisable		Weighted-Average Remaining Life
	Number of Options	Weighted-Average Exercise Price	Number of Options	Weighted-Average Exercise Price	
\$13 - \$15	691,148	\$13.51	456,397	\$13.52	7.4
\$15 - \$17	164,184	\$15.00	164,184	\$15.00	3.6
\$17 - \$19	170,206	\$17.40	160,706	\$17.39	4.6
\$19 - \$21	521,542	\$19.75	511,708	\$19.77	6.5
\$21 - \$22	849,982	\$21.58	263,435	\$21.61	9.0
\$22 - \$24	758,409	\$23.03	300,034	\$22.41	8.4
\$24 - \$26	24,000	\$25.50	—	—	9.9

The 1995 Lincoln Stock Purchase Plan provides employees the ability to purchase open market shares on a commission-free basis up to a limit of ten thousand dollars annually. Under this plan, 400,000 shares have been authorized to be purchased. There were 53,035, 6,271 and 26,559 shares purchased in 2002, 2001 and 2000, respectively.

NOTE E — RATIONALIZATION CHARGES

During the first quarter of 2002, the Company recorded rationalization charges of \$10,468 (\$7,045 after-tax), or \$0.17 per diluted share. The rationalization charges were principally related to a voluntary retirement program affecting approximately 3% of the Company’s U.S. workforce and asset impairment charges. Workforce reduction charges were \$5,353, while non-cash asset impairment charges were \$5,115.

NOTE F — SHORT-TERM AND LONG-TERM DEBT

At December 31, 2002 and 2001, long-term debt consisted of the following:

	2002	2001
Senior Unsecured Notes due 2007, interest at 5.58%	\$ 40,000	\$ —
Senior Unsecured Notes due 2009, interest at 5.89%	30,000	—
Senior Unsecured Notes due 2012, interest at 6.36%	80,000	—
8.73% Senior Note due 2003	9,375	18,750
Foreign borrowings due through 2011, interest at 2.3% to 10.0% (2.6% to 12.4% in 2001)	8,409	10,725
Capital leases	5,991	3,891
Other borrowings due through 2023, interest at 2.0% to 7.9%	13,804	4,249
	187,579	37,615
Less current portion	13,433	13,434
Total	\$174,146	\$24,181

Physical Hand-Delivery

Hand-delivered submissions can be made to the Central Imaging Facility (CIF):

- **Address:** 1 Centre, Street, Room No. 1329, New York, NY 10007
 - 13th Floor
- **Phone:** (212) 669-3741

CIF Hours: Monday-Friday, 9am-5pm (closed on City holidays).



Direct Interface Submissions



- **Hours:** 9am-5pm (for same day processing)
- **Reminders:**
 - Direct Interface submissions after these hours will not be considered received until the following business day for the purposes of the Comptroller's 30-day review.
 - Direct Interface made via PASSPort or a similar system will be validated to confirm that approved procurement methods were selected, and that documents corresponding to mandatory fields were attached.
 - If your submission is rejected due to a validation issue, follow error messages guidance or contact the Comptroller's Bureau of Information Systems and Technology (BIST) at helpdesk@comptroller.nyc.gov.



Alternative Digital File Drop Submissions

When agencies can't submit a contract through an electronic system like PASSPort, or via hard copy, Citrix Sharefile and Microsoft Sharepoint allow them to transmit a contract via a single PDF file. Agency staff use these platforms to deposit a contract file in a shared-folder accessible by the Comptroller's office.

- **Requirements:**
 - Email ocamailbox@comptroller.nyc.gov and ClFrequeest@comptroller.nyc.gov to request a link
 - Ensure the Contract #, OCA #, EPIN, and Vendor information are consistent across all contract documents. These values must match what is in FMS.
 - Contracts can only be submitted as a single PDF



Contract Submission Portal (CSP)

- The CSP is a Comptroller-operated contract submission application (housed in Citrix).
- Unlike with Sharefile, agency staff can use the CSP to label individual contract documents.
- Submissions are automatically routed to BCA.
- Interested in signing up?
Email ocamailbox@comptroller.nyc.gov

The screenshot displays the OAISIS Contract Submission Portal interface. At the top, there is a header bar with the title "OAISIS Contract Submission Portal" and a search bar. Below the header, there are several dropdown menus and input fields for contract details, including "Transaction Code" (CT1), "Contract Number" (07 LESSEE NEGOTIATION), "Contract Number" (20251412526), "OCR Number" (1016532186), "Vendor Number" (V500066459), and "Contract Number" (10220252024271). The main area of the portal is organized into a grid of document upload sections, each with a "Choose File" button and a "No file chosen" status. The sections include: "Advice of Award" (with a file named "aoa.pdf"), "Agency Memo" (with a file named "agm.pdf"), "Agreement" (with a file named "lease.pdf"), "Bid Tabulation", "Budget Detail" (with a file named "agmemo.pdf"), "Contract Cover Sheet" (with a file named "cics.pdf"), "DOI Vendor Name Check Memo", "Emergency Declaration", "Insurance Certificate", "Late Registration Memo", "Miscellaneous", "OMB Approval", "Pre-Solicitation Review Report", "Proposal/Bid", "Recommendation For Award", "Recommendation for Renewal", "Responsibility Determination" (with a file named "w9.pdf"), and "RFP/IFB". Each section also has a small "Add" button next to the "Choose File" button.



An aerial photograph of the New York City skyline, featuring the Chrysler Building and the Manhattan Bridge. The image is overlaid with a semi-transparent blue filter. A dark blue rounded rectangle with a white border is centered in the middle of the image, containing the text "Important Last-Minute Checks" in white.

Important Last-Minute Checks

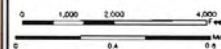
NAVSTA Newport

Commander Navy Region
Mid-Atlantic GeoReadiness
Center

Legend

- Gates
- Base Boundary
- Existing Structures
- Ammunition Storage Area
- Fence Line
- Runway
- Taxiway
- Helipad
- Apron
- Shoulder Overrun
- Aircraft Parking Area
- Railroad
- Golf Course
- Playground
- Swimming Pool
- Athletic Court
- Athletic Field
- Existing Piers
- Drydock

Print Date: 10 Oct, 2021



Coordinate System: NAD83 for UTM Zone 17 N

GeoReadiness Center

AM-OS Mid-Atlantic
Norfolk, VA 23511
(757) 444-3013



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3.0 Applying Best Practice Strategies

State, regional, and local agencies have developed innovative approaches and best practices to deliver projects on time and on budget by preventing and minimizing avoidable change orders or making the most effective use of unavoidable change orders. These can be generally bundled into the following categories:

- Ensuring consistency through documented **processes**
- Employing **risk management**
- Fostering **communication** and internal and external partnering
- Using **alternative contracting** methods
- Developing **workforce** capacity and experience
- Leveraging **data and technology**
- Considering **organizational culture** and barriers to decision-making.

Each of these strategies can be implemented independently or in conjunction with others to improve project delivery. While formalized processes, communication, and risk management can be standalone approaches to address change order concerns, they are often also foundational elements to implementing the other best practices.

For each best practice strategy, this section offers considerations for practitioners and leaders to better understand how they can address change orders within their team, department, or organization.



Figure 1: Best Practice Strategies to Prevent or Minimize Change Orders



3.1 Ensuring Consistency Through Documented Processes

Standardized workflows and procedures help practitioners understand roles, responsibilities, relationships, and expectations in project delivery. Having documented, repeatable processes in place can increase quality, consistency, and predictability as well as facilitate knowledge transfer and organizational continuity. As programs mature, periodic assessment of processes can ensure they serve their intended purpose and reflect lessons learned.

3.1.1 Process Considerations

Project Agreements and Planning

Processes should be built to ensure that the project agreement is only executed once the project scope is fully defined and the design is verified. The plans and specifications contained in the project agreement are the project's "blueprint" that reduces the need for change orders. Standards should clearly delineate quality, systems, and responsibility requirements, but not be so prescriptive as to induce extraneous costs. Contract development processes should account for early legal counsel and sufficient time for quality reviews to identify and eliminate ambiguities, errors, and material omissions. Project agreements should provide mechanisms to navigate potential cost and schedule escalations and consider accountability measures that provide cost recovery for certain causes (e.g., design errors and omissions).

Processes that result in a strong project agreement minimize change orders by incorporating high-quality work products from project development and clearly outlining responsibilities, while also providing mechanisms to efficiently navigate change orders that are unavoidable.

Explore construction contract products featured on the [U.S. DOT Project Delivery Center of Excellence website](#).



Estimating and Scheduling

Budget and schedule estimates should rely on realistic and risk-based contingency and adjustment factors to account for potential changes in designs, scopes or cost estimates. Realistic estimates and schedules that are vetted and developed through consistent processes can reduce the likelihood or adverse impacts of change orders.

See [Data and Technology](#) and [Risk Management](#) best practice strategies for more information.



Procurement

Competitive, rigorous, and transparent procurement processes complement project agreements. These processes can incorporate industry engagement through events where project sponsors provide presentations, engage in questions and answer sessions, or participate in one-on-one discussions to give valuable feedback on how to structure contracts and procurements. Detailed Requests for Information (RFIs), Requests for Qualifications (RFQs), and Requests for Proposals (RFPs) identify material contractual terms and conditions; elicit thoughtful and complete proposals from qualified bidders; and discourage

underbidding that exploits change orders, renegotiations, or the filing of claims for later financial gain.

See [Communication](#) and [Alternative Contracting Methods](#) best practice strategies for more information.



Subsurface Investigations and Utilities

Comprehensive review of underground conditions and early coordination with relevant third parties (i.e., utility, railroad, ROW owners) reduces the need to reroute or redesign projects due to conflicts discovered late in the construction phase, saving time and costs. Establish checkpoints throughout project development to keep the project team informed of potential conflicts that could result in change orders.

See [Communication](#) best practice strategy for more information.



Addressing and Negotiating Change Orders Efficiently in Construction

The project agreement must clearly outline the change order process. Robust communication of this process is also critical to ensure project teams are prepared to efficiently manage and negotiate change orders when they do occur.

Process elements to consider include how to identify, document, and navigate:

- The need for a change order
- Which stakeholders must participate in the process
- Terms and price negotiation
- Approval authorities
- Deadlines for escalation
- Dispute resolution.

Addressing Federal Requirements

When projects use federal funding, even as a portion of the investment, U.S. DOT agencies—such as the Federal Aviation Administration (FAA), Federal Highway Administration (FHWA), Federal Railroad Administration (FRA), Federal Transit Administration (FTA), Maritime Administration (MARAD), and Pipeline and Hazardous Materials Administration (PHMSA)—perform oversight and develop guidance for compliance and efficient delivery of federal

funding. Ensure your organization has procedures in place that align with the funding provider's requirements to ensure a change order will not impact a project's funding eligibility.

Explore the [Project Delivery Toolbox's](#) federal oversight resources.



Maintaining Standard Operating Procedures (SOPs)

SOPs help to reduce reliance on tacit knowledge or ad-hoc, inconsistent processes and decision-making to improve continuity. They provide a formal reference of processes and roles, helping to enforce accountability, enhance oversight, and reduce mistakes and omissions. They also help to optimize effective use of contractors. SOPs that are thorough and well understood by the project team can help guide sound decision-making that prevents the need for change orders.

See [Communication](#), [Workforce](#), and [Organizational Culture](#) best practice strategies for more information.



3.2 Employing Risk Management

Deliberate risk management helps to improve decision-making across project activities and minimize or mitigate potential change orders. Risk management provides a structured way to navigate the array of uncertainties, variabilities, opportunities, and threats that may affect a project's ability to achieve its objectives¹. Risk management helps agencies understand and plan for uncertainties or overall risk through systematic application of strategic tools.

A key step in risk management is documenting the responsibility for risks in project agreements. Developing a robust project agreement involves not only assessing risks but also assigning responsibility for them within the contract. The project agreement enables project sponsors to accept, mitigate, transfer, avoid, or enhance areas of uncertainty, effectively managing risks. Assigning risks creates a mutual understanding of liability for response and mitigation measures before problems arise, helping to reduce the adverse impacts of change orders. Liability for risks generally should be assumed by the party most able to manage them. Inappropriate risk shifting can lead to unnecessary change orders, disputes, or increased costs for contractors to cover them. Actively identifying and managing risk can reduce the uncertainty that causes change orders.

¹ https://www.planning.dot.gov/documents/FHWRiskMgmtRoadmap_July2018.pdf

3.2.1 Risk Management Considerations

Risk Management Process

Risk management provides a systematic approach to minimizing threats and maximizing opportunities to make decisions that will determine project success. It is essential to develop a comprehensive project agreement. While difficult to account for every possible risk a project may face, the risk management process provides a framework and can be a helpful tool to plan for instances when unknown risks occur later in the project. Generally, steps include identifying, assessing, planning, monitoring, and controlling risks. This process can be applied throughout each phase of the project lifecycle and at program and organizational levels. Early identification of risks allows project stakeholders time to resolve issues that could lead to change orders before projects get to construction.

See [Process](#) best practice strategy for more information.



Allocating Risk Based on the Project

Properly allocating risk between the agency and the contractor is important to containing costs. Consider different pricing models and contracting methods to carefully balance the risks being assigned to each side. Not every project type is suited for one project delivery method. Evaluating the most appropriate project delivery method for individual projects or project types can reduce risk. Clearly defining which party owns project risks reduces misunderstandings and unpredictability that can lead to the need to amend construction contracts through change orders.

See [Alternative Contracting Methods](#) best practice strategy for more information.



Considering Risk in Budget and Schedule Estimates

Probabilistic risk-based estimating (PRBE) methods provide decision makers with reliability-based information about uncertainty around project objectives. PRBE is a scalable technique that uses simulation to develop distributions of cost and schedule completion. Distributions provide the entire range of possible outcomes, exposing areas of uncertainty typically hidden in conventional, deterministic estimates.² Cost related change orders are less likely to be necessary when project sponsors have a high degree of understanding and confidence in project estimates.

² [FHWA/NHI PRBE for Highway Project Cost and Schedule \(FHWA-NHI-134205\)](#)

Enterprise Risk Management (ERM)

ERM assesses risk to enhance the overall success of an agency rather than focusing solely on individual business units, programs, or projects. It strategically addresses risk by aligning goals across organizational levels. Executives play a key role in championing this process and fostering a culture of risk management. Organizational uncertainties could include reputation, data integrity, funding, safety, and policy. An organizationally directed culture of risk management is integral to deploying project teams prepared to mitigate the risks in project delivery that can result in change orders.

See [Organizational Culture](#) best practice strategy for more information.



3.3 Communication and Partnering

Change orders may originate from poor communication throughout a project's lifecycle. Early and frequent communication, partnering, and relationship building strengthen teams and produce information and solutions that can prevent project changes. Incorporating communication checkpoints into project delivery processes can help bridge information gaps to solve problems before they become challenges that require change orders to resolve. Early involvement of project stakeholders, understanding of the project agreement by all parties to it, and mutual efforts to foster collaborative relationships all result in benefits to successful project delivery.

Open and honest communication, whether integrated into formal processes or outlined in the project agreement, is helpful to both internal and external partners like project stakeholders, consultants, and the contracting community. Strong communication is embodied by teams that question preconceptions and consider project circumstances from different viewpoints. Partnering can manifest through informal or formal communication processes to create strong working relationships that reduced the number and impact of change orders by allowing space to the needs of all project stakeholders.

3.3.1 Communication Considerations

Internal and External Communication

Frequent communication across functional units and external partners and at key milestones can help to address some of the causes of construction change orders early in the planning, design, and procurement phases. Effective internal communication can promote alignment on the resolution of critical issues that may otherwise result in change orders or escalate to claims.

See [Process](#) best practice strategy for more information.



Early Contractor Involvement

Early contractor involvement can benefit the project by improving transparency, leveraging contractor experience, while identifying and addressing risks collaboratively to avoid issues later in the project lifecycle. Solicit industry feedback through RFIs, pre-solicitation conferences, or alternative contracting methods to increase understanding of project context, needs, and requirements. Soliciting industry expertise and treating well-informed contractors as partners early in the project development process can contribute to more thorough plans and designs that are less vulnerable to change orders.

See [Alternative Contracting Methods](#) best practice strategy for more information.



Right of Way and Permitting

Early coordination with entities that own assets on or near your project site or that issue permits can avoid delays during construction. Formalize solutions through pre-award railroad, utility, or permitting agreements. Frequent and good-faith communication with these entities may promote programmatic solutions that prevent conflicts that cause change orders on future projects.

See [Process](#) best practice strategy for more information.



Culture of Communication

Clear and open communication between agency leadership and technical staff is critical to support effective decision-making that precludes the need for avoidable change orders during construction. An organizational culture that establishes safety for practitioners to critically examine issues through open communication can improve project delivery practices at all levels. Importantly, it empowers project staff to speak up about issues before they can manifest as change orders.

See [Organizational Culture](#) best practice strategy for more information.



Stakeholder and Public Outreach

Early engagement and collaboration with stakeholders and the public to identify mutual, complementary goals and potential issues helps to identify solutions and avoid change orders late in project delivery.

Involving Industry

Actively engaging in collaborative problem-solving with industry partners can foster stronger relationships and avoid adversarial situations that might result in unproductive use of change orders that increase costs and cause delays. Understanding the constraints of the contracting community and staying abreast of evolving industry trends helps to develop stronger project agreements, inform process improvements, and better prepare for potential change orders.

Formal partnering efforts with industry can include documented agreements, roles, and procedures. Informal partnering and relationship-building opportunities happen at project meetings and industry summits and during post-construction reviews. Incorporating industry perspective into project development can help agencies address issues that might otherwise result in change orders before they can impact a project.



3.4 Alternative Contracting Methods

Employing alternative contracting methods balances agency and contractor perspectives and risks to minimize project vulnerabilities that can lead to change orders. Transportation infrastructure construction contracts have historically been governed by low-bid, design-bid-build (DBB) contracting methods where the project sponsor is responsible for designing the project and awarding the contract to the lowest responsible bidder, while the contractor is responsible for constructing the project according to the terms established in the project agreement.

While alternative contracting is not appropriate for all project contexts, project sponsors may consider, as appropriate and permitted by state and local law, methods like best value selection, design-build, progressive design-build, construction manager/general contractor, public-private partnerships, and alliance contracting to create opportunities for early contractor involvement and collaboration. Under such project agreements, project sponsors and contractors are also able to collaboratively determine the most efficient and fair allocation of risk, which can minimize change orders which result in unexpected cost and schedule growth during project construction.

Other mechanisms such as incentives and disincentives or value engineering processes provide sponsors with options for flexibility, innovation, and accountability within conventional contracting frameworks.

3.4.1 Alternative Contracting Considerations

Selecting a Project Delivery Method

The use of alternative delivery methods requires enabling authority in federal, state, or local legislation. A decision-making process should be in place to determine the appropriate approach for a given project based on goals, risks, pricing model, assets, capacity, stakeholder needs, and experience. Decision matrices are a useful tool to objectively weigh factors to select a legally authorized contracting method. Project sponsors may consider different methods based on:

1. Cost impacts
2. Schedule impacts
3. Opportunity to manage risk
4. Complexity of design and construction phasing
5. Stakeholder engagement
6. Opportunity for innovation.

FHWA's Every Day Counts Initiatives and Special Experimental Project No. 14 - Alternative Contracting provide examples of how some agencies have investigated the use of alternative contracting methods and the reasoning behind selecting a particular method, which can include reducing change orders.

See [Process](#) and [Risk Management](#) best practice strategies for more information.



Encouraging Contract Compliance

Incentive and disincentive structures, pricing models, and bid options may encourage successful cost and schedule performance as well as compliance with contractual obligations by making the initiation of change orders or claims disadvantageous to all parties of the project. Alternative contracting methods introduce new ways to involve the contracting community in project development and negotiating the terms of project agreements. Adherence to a project agreement's terms and conditions minimizes the number of disputes that may necessitate change orders.

Organizational Capacity to Administer Alternative Delivery Methods

Ensuring the organization has the necessary resources and expertise to administer a project using the proposed delivery method helps to minimize or mitigate risks that may result in change orders during construction. Alternative delivery methods often require project sponsors to relinquish some control over risk tolerance, costs, and decision-making. Internal staff should be well-versed and encouraged to lead a project through a more flexible

environment and still achieve the desired results. This may require new procedures, standards, and specifications to advance projects with less robust designs and accommodate contractor-driven innovation and methods.

See [Workforce](#) and [Process](#) best practice strategies for more information.



3.5 Workforce Capacity and Experience

Access to a robust and skilled workforce is essential to manage projects and produce complete, consistent, and quality work that minimizes the need for change orders. An insufficient or inexperienced workforce may lead to change orders. Agencies need sufficient and competent staff with the requisite legal, operational, and technical expertise to manage the development and execution of the project agreement.

3.5.1 Workforce Considerations

Technical Training and Development

Training opportunities are available to support agency workers, including Local Technical Assistance Programs, Tribal Technical Assistance Programs, Rural Transit Assistance Programs, and University Transportation Centers. Engaging with industry associations and their regional counterparts and institutes such as the American Association of State Highway and Transportation Officials (AASHTO), American Council of Engineering Companies (ACEC), Associated General Contractors of America (AGC), American Public Transportation Association (APTA), American Public Works Association (APWA), American Road and Transportation Builders Association (ARTBA), American Society of Civil Engineers (ASCE), National Association of County Engineers (NACE), and National Association of City Transportation Officials (NACTO), among others, can provide learning opportunities and access to training products. Federal training resources include the National Highway Institute, National Transit Institute, and Federal Aid Essentials. Utilizing these training resources can help agencies achieve a workforce that is knowledgeable about the risks and challenges that contribute to change orders on transportation construction projects.

Documenting Lessons Learned

Conducting post-construction reviews and documenting lessons learned are valuable processes to build and maintain institutional knowledge of project development and delivery. Technical staff can organize knowledge capture programs within and across teams for an organic approach to training to prevent knowledge gaps that later cause change orders, even as staff turnover occurs.

Project Management Training

Though not all project staff need to be as well-versed as the project managers, some amount of training in project management can improve performance in management of individual work streams, contractor progress, and project controls. Transparency and attention to the progression of project activities by all contributors can help to avoid oversights and pitfalls that can later require change orders to resolve.

Developing Organizational Capacity

Cross-training agency staff across roles and subject matter expertise can improve an agency's capacity to administer projects of varying scopes and complexities. If consultants are expected to be members of project delivery teams, procurement through qualifications-based selection and inclusion of the consultant community through partnering and training alongside internal staff can contribute to developing a culture of quality and consistency. The occurrence of change orders can be reduced as an agency's workforce knowledge and capacity increases to lower the incidence of incomplete or incorrect work products.

See [Process](#) best practice strategy for more information.



3.6 Data and Technology

Data and technology support more efficient decision-making and problem solving that reduces negative change order outcomes by facilitating communication, combating bias, and identifying risk areas. Data collection and evaluation help to identify recurring or emerging risk areas, enabling allocation of resources to the most impactful cost and schedule solutions. They also provide an opportunity to institutionalize project lessons learned to improve future delivery. Data can also mitigate internal and external biases and misrepresentations by supporting decision-making with objective facts.

Technology tools such as Building Information Modeling (BIM), Advanced Digital Construction Management Software (ADCMS), Digital As-Built, E-ticketing, Performance Dashboards, and risk simulators enable project sponsors to leverage data more easily and develop solutions earlier.

3.6.1 Data and Technology Considerations

Collect and Manage Change Order Data

Documenting the specific causes of change orders helps to identify patterns and develop metrics to assess the frequency and impact of change orders, enhancing determination of whether actions or corrections are necessary. Using data to evaluate past occurrences and

trends is a useful first step in identifying and evaluating change order risks that could jeopardize the efficiency of projects and programs. This data can be used to develop better informed estimates and risk assessments for different project types based on the type of asset or construction method (also referred to as “reference class forecasting”). ADCMS solutions can provide further streamlining of such analyses by enabling staff to assign codes to change order causes that can be used to aggregate reports that demonstrate trends. Tracking performance outcomes helps to improve the decision-making process by identifying where to allocate organizational resources to reduce the likelihood and impact of change orders that pose the most risk to a project’s cost and schedule.

See [Risk Management](#) best practice strategy for more information.



Utilizing Data and Technology in Design, Construction, and Reporting

Technology solutions can improve the consistency and communication of project and asset information, allowing project team members to identify and address issues that may lead to change orders before construction begins. When using data management processes and technology, regular reviews are important to determine whether existing tools are meeting organizational needs or if capacity needs to be expanded.

BIM and 3D modeling can promote design quality by facilitating early identification of conflicts. Information tools can support communication with project sponsors through easily digestible visual models, helping to avoid changes from misunderstandings.

Digital as-builts identify and monitor conflicts with existing conditions and facilitate partnering with utility and other asset owners to ensure conflicts are identified and addressed timely.

ADCMS streamlines interoperability, project management, submittal processes, communication, contraction administration, and approval processes to facilitate the execution of necessary change orders and accelerate communication that highlights potential issues that can become change orders.

E-ticketing monitors real-time use of project quantities to anticipate the potential need for adjustments or changes.

Dashboards facilitate performance monitoring, evaluation, transparency, and accountability to more quickly identify project issues that can result in change orders and encourage project sponsors to deliver projects that maintain the integrity of promised schedules and costs.



3.7 Organizational Culture

Organizational culture sets the tone for project delivery by aligning goals, defining responsibilities, and empowering the workforce to create a healthy project ecosystem that is prepared to address change orders and their causes. It permeates all aspects of project delivery, and as such has the potential to contribute to change orders downstream.

Leadership is responsible for establishing organizational values around quality and successful project delivery, advocating for change, and ensuring accountability to meeting goals. Creating a culture of alignment allows for candid communication and dissenting views without the threat of retaliation or adversarial pressures. Therefore, leaders should be chosen based on experience and demonstrated success in exercising those actions.

Project sponsors have distinct functions, divisions, and teams, and projects change hands multiple times throughout development. These hand-offs can result in changes³ or delays when teams work in isolation and are not informed of priorities or involved in each other's efforts. An organizational culture that centers collaboration amongst teams and the safety of participants to raise issues is essential to identifying and addressing project risks and weaknesses before they become major issues requiring change orders.

Creating a "project first" culture aligns project goals and priorities across leadership, technical teams, and stakeholders to ensure execution of the vision without unnecessary cost overruns, duplication, and delays.

3.7.1 Organizational Culture Considerations

Multidisciplinary Project Teams

Cross-functional teams can help projects advance more efficiently and ensure goals and priorities remain consistent, and decisions are made holistically to avoid project gaps that can later result in detrimental change orders. Creating opportunities for interdisciplinary collaboration can also result in value engineering proposals or innovations.

See [Communication](#) strategy for more information.



³ [NACTO Structured for Success](#)

Organizational Structure

Aligning funding and staff with clearly defined roles and responsibilities makes it easier to agree on priorities and implement projects. Mapping and documenting processes can help establish more transparent norms and practices, reducing redundancies and improving communication and handoffs between teams to prevent project development oversights that can later require change orders to rectify.

See [Process](#) best practice strategy for more information.



Putting the Project First

Clear priorities and shared goals between project parties are essential for successful projects. A project-first culture that prioritizes project quality empowers project staff to dedicate sufficient time to develop realistic project deliverables and incorporate stakeholder feedback. Similarly, projects benefit when each involved party understands and appreciates the others' goals early in project development. This helps to prevent rushing projects to move through programmed funds, which can lead to downstream cost and schedule consequences.

3.8 Chapter Summary

There are many approaches to preventing and minimizing the likelihood and impact of construction change orders. The best practice strategies presented in this chapter can be used independently or in combination with each other to address technical, organizational, financial and funding causes of change orders depending on the pervasiveness of change orders in a program. Practitioners can consider how to:

- Implement formal processes that ensure quality, consistency, and completeness of technical work products
- Use risk management to improve decision-making that targets project elements that can potentially cause change orders
- Promote early and frequent communication with internal and external stakeholders to fortify project teams and facilitate information exchange as well as the understanding of stakeholder roles and responsibilities in the project delivery lifecycle to ensure project designs and agreements are completed with input from all perspectives
- Employ alternative contracting methods where appropriate to balance project sponsor and contractor risks and minimize project vulnerabilities that can lead to change orders
- Access a robust and skilled workforce that has the capacity to manage projects and produce work free from errors, omissions, and oversights that require change orders to correct

- Leverage data and technology to support more efficient communication, decision-making, and problem solving that precludes the need for construction change orders
- Nurture an organizational culture that establishes a project-first environment where stakeholder goals are aligned and respected and practitioners are empowered to think critically.

These strategies help practitioners strengthen project agreements to ensure the reasons for change orders are considered and addressed to make projects resilient to the impacts.

4.0 Noteworthy Practices

Noteworthy practices demonstrate how sponsors of federally funded projects have successfully implemented variations of the best practice strategies to address the impacts of change orders. These stories illustrate that change orders can arise from various issues and be addressed through a variety of best practice strategy solutions. When organizations earnestly assess opportunities to improve their practices and prioritize communication in their processes, many benefits, including reducing or better managing change orders, can be realized. The following examples are multidimensional, applicable across modes, and scalable to fit practitioner needs.

- Washington State DOT (WSDOT) deploys a probabilistic risk-based estimating (PRBE) approach to cost and schedule estimating that helps to ensure sufficient accounting of project risks to reflect the most realistic range of costs and schedule implications. WSDOT also developed the Cost Estimate Validation Process for high-cost projects. Together, these position WSDOT to avoid or reduce change orders impacts due to inaccurate estimates on their projects.
- Colorado DOT uses a Subsurface Utility Engineering (SUE) program to communicate with utility owners to manage subsurface utility information throughout the project lifecycle. This contributes to fewer change orders, delays, and safety hazards resulting from utility conflicts by providing the project team with more complete and accurate information to develop design and project agreement documents.
- Texas DOT's Design and Professional Engineering Procurement Services divisions developed new procedures that partner with the consultant community to improve design quality, consistency, and accountability. These processes increase workforce capacity and help prevent costly and time-consuming change orders required to address design errors and omissions discovered during the construction phase.
- The San Diego Association of Governments and Valley Metro of Phoenix, AZ use alternative contracting methods to appropriately distribute project risks between parties. By proactively addressing risk collaboratively with the contracting community, project sponsors are able to resolve design and constructability concerns early in project development and formalize responsibilities through a comprehensive project agreement, to prevent challenges that can later become change orders without adversely impacting agencies or contractors.
- North Carolina DOT uses constructability reviews and post-construction assessments to improve communication between design and construction teams before and after projects are constructed. This contributes to a better working relationship between the DOT and contractors, while building the capacity of the workforce to better identify and prevent change orders on future projects.

- Montana DOT employs its Partnering Program to ensure early and continuing coordination among project partners throughout the project development and delivery phases. The program produces better working relationships between the DOT and contracting community to collaboratively address risk areas that can necessitate change orders. It also implements a structured approach to problem resolution, reducing change orders and preventing issues from escalating into a claim or litigation.

Noteworthy Practice: Washington Department of Transportation (WSDOT) Cost Estimate Validation Process

One of the most difficult aspects of project delivery is sufficiently accounting for risk when developing cost and schedule estimates. Unaccounted risk frequently leads to change orders and other cost and schedule overruns. Probabilistic risk-based estimating (PRBE) is a scalable approach to cost and schedule estimating that helps to ensure sufficient accounting of project risks. PRBE uses data and statistical simulation to identify potential risks before significant design decisions are made or construction begins. Good processes and organizational culture are foundational to PRBE. The goal of PRBE is to identify the risks to a project, quantify their potential cost and schedule impacts, and determine the best methods for managing and communicating them.

WSDOT has used PRBE methods for over two decades to successfully deliver hundreds of transportation infrastructure projects.

Change Order Issue

Traditional cost estimating may insufficiently account for project-specific risks, obscure flaws in the estimate or design, or underestimate the degree of uncertainty within the estimate.⁴ WSDOT historically used this method of estimating, which relied on a project estimator's experience and judgement without explicitly identifying a project's individual uncertainties and risks.⁵

During the 1990s, WSDOT faced public mistrust of their ability to manage projects and communicate estimates. In particular, WSDOT's State Route 167 expansion project saw compounding cost and schedule growth that far exceeded the original estimate.⁶ WSDOT needed a PRBE process to help them better identify and account for the "unknown-knowns" and "unknown-unknowns" that make it difficult to estimate project costs and schedules.

PRBE Risk Assessments

Qualitative risk assessment addresses subjective risk elements and may be sufficient for small projects.

Quantitative risk assessment numerically evaluates risk elements and may be necessary on more complex projects. It involves techniques such as **Monte Carlo** simulation (WSDOT Project Risk Management Guide).

Monte Carlo simulation computes probability to determine a distribution of cost or schedule outcomes (NHI, PRBE for Highway Project Cost and Schedule).

⁴ Gabel, M., Sujka, M., Davis, Z. W., & Keizur, A. E. (2023). Performance of Risk-Based Estimating for Capital Projects. Transportation Research Record, 2677(1), 1059-1070. <https://doi.org/10.1177/03611981221103238>

⁵ [WSDOT Project Risk Management Guide, P.1-7](#)

⁶ [Building Public Trust | FHWA \(dot.gov\)](#)

Best Practice Solution In 2002, WSDOT developed the Cost Estimate Validation Process (CEVP®) for high-cost projects. This PRBE process assesses risk using historic qualitative and quantitative data to produce estimates expressed as a range of values, rather than an individual number. This value range aims to convey the degree of uncertainty in the estimate, which can assist the agency in project-specific risk management. WSDOT creates two components for estimates—the base cost (or likely cost) component and the risk component.⁷ The risk component includes defined events, with each having a corresponding likelihood and consequence to produce a risk register. The estimate is then subject to a robust review by independent subject matter experts for validation of the assessment.

What is a risk register? A risk register is a control document for risk identification, risk updates, and risk treatment actions. It records all identified risks and includes details and characteristics for those risks (NHI, PRBE for Highway Project Cost and Schedule).

WSDOT has continually refined the process for use on projects of varying sizes and risk profiles. After initial success with the CEVP for large projects, WSDOT began using a scaled-down version, referred to as Cost Risk Assessment (CRA) workshops, for smaller and less complex projects.

In addition to improving decision-making and project outcomes, WSDOT uses the results of CEVP/CRA workshops to communicate project risks to the public and project partners through simple and concise one-page summaries. These not only help to improve accountability, but also appropriately establish expectations and perceptions of cost growth.

State and agency leadership gave early support to the development and implementation of PRBE. Adopting the practice was an organizational effort made easier by early successes, as well as active and continuous improvement efforts.⁸ WSDOT has documented lessons learned and provides targeted training to continue to build an organizational culture of risk management.⁹

Involvement in PRBE is organization wide, including executives, senior leaders, and project staff from across departmental units. External consultants and subject matter experts may also participate in the workshops to stymie bias and assist with validating estimates.¹⁰

Major Considerations

WSDOT embedded risk-management into their organizational culture and existing processes, which can be a challenging but important undertaking for any agency. Agencies looking to

⁷ [WSDOT Project Risk Management Guide, P.1-7](#)

⁸ Gabel, M., Sujka, M., Davis, Z. W., & Keizur, A. E. (2023). Performance of Risk-Based Estimating for Capital Projects. Transportation Research Record, 2677(1), 1059-1070. <https://doi.org/10.1177/03611981221103238>

⁹ [WSDOT Project Management training webpage](#)

¹⁰ [WSDOT Project Risk Management Guide, P.A-1](#)

implement or strengthen their risk-based estimating processes should develop strategies to engage with leadership and cultivate their support. Once implemented, maintaining and improving the risk-based estimating process is critical to its success.

Resources can be a constraint for many agencies looking to implement more robust risk-based processes. To address resource constraints, such as internal staffing availability, available technology, and budgets, WSDOT scales their PRBE process based on project size or complexity. Agencies may need to evaluate their staffing levels, budgets, and project profiles to implement appropriately scaled processes.

Results

WSDOT's implementation of PRBE has been a success in achieving better project outcomes and organizational change.¹¹ WSDOT projects that used a CRA process have shown a reduction in cost and schedule growth related to change orders, compared to projects that did not use a CRA process. On average, this reduction is 4.8 percent for cost and 6.7 percent for schedule.¹² Consistently delivering projects on time and on budget to build public trust was one of the main goals of WSDOT's CEVP. It has also resulted in building an organizational culture of risk management supported by [dozens of resources](#) and a robust [Project Risk Management Guide](#) to consistently implement their processes.

Takeaways

Agencies of any size can use PRBE to mitigate cost and schedule growth, communicate valid and realistic expectations to stakeholders, and more efficiently allocate resources. Potential strategies include:

- Implement a standard project risk assessment process:
 - Collect data to document historical risk likelihood and impacts
 - Conduct risk assessment workshops
 - Use data to inform and validate reference class forecasting
 - Escalate levels of analysis based on project profiles (size, complexity, etc.).
- Develop materials to communicate risk with project teams and external stakeholders.
- Record best practices and lessons learned to build organizational culture.

For more information, contact WSDOT's CRA and CEVP program manager at:
Mark.Gabel@wsdot.wa.gov.

¹¹ Davis, Z., Gabel, M., Sujka M., & Keizur, A. E. (2022). [Implementing and Sustaining a Cost Risk Estimating and Management Program](#). International Conference on Transportation and Development.

¹² Gabel, M., Sujka, M., Davis, Z. W., & Keizur, A. E. (2023). Performance of Risk-Based Estimating for Capital Projects. Transportation Research Record, 2677(1), 1059-1070. <https://doi.org/10.1177/03611981221103238>

Noteworthy Practice: Colorado Department of Transportation (CDOT) Subsurface Utility Engineering

Utility conflicts are a common cause of change orders on transportation construction projects. Subsurface utility engineering (SUE) is the process of obtaining, identifying the quality of, and managing subsurface utility information throughout the project lifecycle.¹³ Establishing a SUE process can involve overcoming organizational barriers to avoid conflicts that can lead to change orders, delays, disruption to the public, and hazards by improving communication and coordination between project partners. Effective SUE relies on good processes to govern its use, which can help agencies manage risk with appropriate data management and technologies.

CDOT has been a leader in developing practices that improve project delivery across the transportation systems they manage, including highway, bridge, transit, rail, and aviation. CDOT's SUE program¹⁴ is one such notable initiative that has seen rapid and significant improvement in recent years.

Change Order Issue

Dated, inconsistent, or unavailable utility information was adding risk to CDOT projects, often resulting in higher upfront project costs as well as change orders.

CDOT also recognized the need for improved SUE processes following a deadly gas explosion in the state, which the National Transportation Safety Review Board attributed in part to a lack of utility documentation.¹⁵

Best Practice Solution

CDOT, the contracting community, state legislature, utility owners, and other stakeholders worked together to develop a bill to address utility risks on infrastructure projects. CDOT's SUE process follows the requirements set forth by Colorado Senate Bill 18-167 and Revised Statute 9-1.5. The revised statute establishes a process for coordination throughout the project lifecycle to assess data and avoid conflicts before they result in change orders or hazards including:

- Notification of CO 811, the communication link between the utility/facility owners and excavators and homeowners, on SUE projects to receive relevant records and data
- Requirement for new utilities to be electronically locatable

¹³ <https://www.fhwa.dot.gov/programadmin/sueindex.cfm>

¹⁴ <https://ops.colorado.gov/sites/ops/files/documents/BestPracticeSubsurfaceUtilityEngineering.pdf>

¹⁵ [NTSB Pipeline Accident Brief, April 17, 2017](#)

- Creation of the Underground Damage Prevention Safety Commission for enforcement and the development of best practices.

American Society of Civil Engineers (ASCE) 38, *Standard Guideline for Investigating and Documenting Existing Utilities* was legally adopted as the means to implement the revised statute's requirements. ASCE 38 provides guidance for performing utility investigations and documenting the results to understand potential utility conflicts and minimize risk by using four levels of SUE data quality—increasing in data precision from Quality Level D (most basic) to Quality Level A (most precise).¹⁶

CDOT bolstered their implementation of ASCE 38 with other measures, including the use of ASCE 75-22, *Standard Guideline for Recording and Exchanging Utility Infrastructure Data*; documented best practices; standardized internal processes; and technology to assist with data management and sharing. ASCE 75-22 complements ASCE 38 by standardizing the collection and exchange of utility data for reliability in digital or non-digital platforms to improve design processes and communications.¹⁷

CDOT leverages a suite of technologies for their improved 3D accuracy, data collection, classification, and management processes. This includes a centralized database that can be used with Geographic Information Services (GIS) and Computer Aided-Design and Drafting (CADD) software. Hardware that collects geospatial data integrates with software compatible with smart phones, tablets, and office computers to allow real-time access, editing, and visualization of the database for project staff.¹⁸ This can help to improve communication and decision-making among project staff wherever they are located.

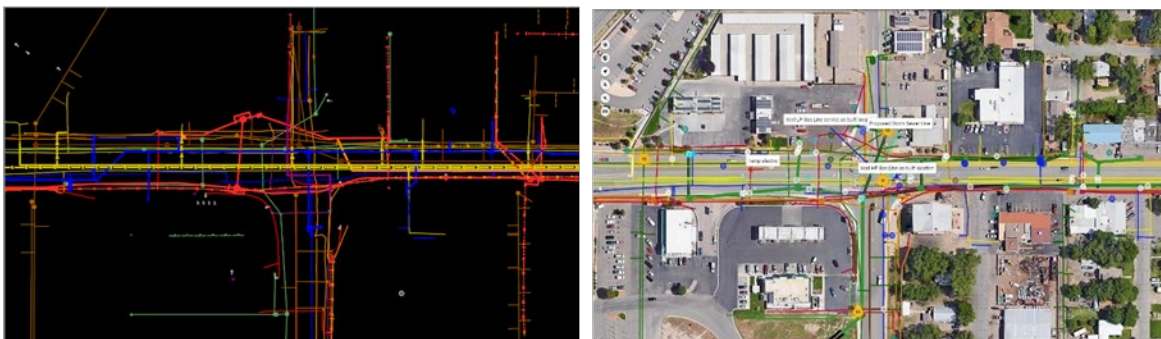


Figure 2: Examples of Utility Data Layers in CADD and GIS.

¹⁶ [Standard Guideline for Investigating and Documenting Existing Utilities \(38-22\) \(asce.org\)](https://www.asce.org/standards-and-guidelines/38-22)

¹⁷ [Standard Guideline for Recording and Exchanging Utility Infrastructure Data \(75-22\) \(asce.org\)](https://www.asce.org/standards-and-guidelines/75-22)

¹⁸ [CDOT-Whitepaper-REV02062023-WEB.pdf \(pointman.com\)](https://www.pointman.com/CDOT-Whitepaper-REV02062023-WEB.pdf)

Major Considerations

Agencies should consider their program's risk profile, industry relationships, and legal environment before revising their SUE program. Legal solutions can yield benefits, but all stakeholders should be engaged in determining which standards and processes will be most balanced, beneficial, and enforceable. Implementing standards and processes through legislation may not be necessary, but it can help strengthen adherence and consistency, as it did in Colorado.

Technological solutions can improve the effectiveness and efficiency of a program or project, but they also can be cost prohibitive for agencies, with or without existing data management systems. Before making investments, agencies should evaluate their needs to determine the type and scale of technologies that may be feasible for adoption.

While particularly helpful for agencies trying to reduce change orders related to utility conflicts, CDOT's SUE program presents a noteworthy example of how creating new processes or retooling existing processes with a risk-based perspective can improve a program. Establishing robust processes and standards to address risks is essential for adopting advanced technologies that enhance direct communication of data and decision-making among project stakeholders throughout the project lifecycle. Improved processes for documentation and data management standards, independent of investments in technology, may also yield benefits in better risk management and communications.

Results

Although use of SUE has been linked to cost savings and fewer change orders for decades,¹⁹ CDOT is still determining the extent to which their SUE program has mitigated potential change orders and excess costs across their program.²⁰ The Central 70 Project was recently completed with zero change orders related to utility conditions. This 10-mile reconstruction of Interstate 70 in an urban context involved removing a 54-year-old viaduct, lowering the interstate, and constructing a 4-acre park above.²¹

¹⁹ Lew, J.J. (2000). [Cost Savings on Highway Projects Utilizing Subsurface Utility Engineering](#)

²⁰ National Academies of Sciences, Engineering, and Medicine. 2022. Implementation of Subsurface Utility Engineering for Highway Design and Construction. Washington, DC: The National Academies Press. <https://doi.org/10.17226/26588>.

²¹ [Central 70 Project Snapshot \(2019\) \(codot.gov\)](#)

Takeaways

SUE encompasses several principles that are fundamentally tied to reducing or mitigating change orders. Agencies should consider:

- Implementation of consistent processes with organizational support
- Standards for collecting and maintaining data, including utilities
- Project screening for specific risks, such as utilities
- Internal and external collaboration to identify solutions
- Formalized communication channels and requirements
- Use of appropriate technological solutions such as software platforms or hardware.

For more information, contact CDOT's SUE program manager at: Rob.Martindale@state.co.us.

Noteworthy Practice: Texas Department of Transportation (TxDOT) Errors and Omissions Process

Change orders and delays are often due to unnoticed or unaddressed errors and omissions (E&O) in contract documents and design plans, specifications, and estimates (PS&E). E&O can result from miscommunication and human error during the stages of project development. Agencies can minimize E&O by implementing processes that encourage quality control of design and contract documents and preparing for mitigation during construction. Assigning responsibility for the E&O and determining the most efficient solution require an organizational culture of coordination and accountability, achievable with open communication and a well-trained workforce able to exercise independent judgement that aligns with established processes. Data can be used to track and identify E&O causes and trends to improve project and programmatic performance.

TxDOT successfully manages thousands of active consultant contracts valued over \$1 billion each year to deliver transportation construction projects across a sweeping multimodal system.

Change Order Issue

From 2001 to 2015, TxDOT determined that approximately one-fourth of all change orders and approximately one-third of all change order value were related to E&O. Following the Federal Highway Administration (FHWA) and internal audits, TxDOT set a goal to reduce the amount and cost of E&O, establish consistency across the organization, and maintain positive relationships with industry.

Best Practice Solution

TxDOT's Design and Professional Engineering Procurement Services (PEPS) divisions published procedures and guidance for TxDOT staff in 2008. This guidance outlines a process to fulfill state law and minimize negative impacts of E&O on projects.²² The process includes several important considerations for preventing and addressing E&O, such as:

- Utilize quality control reviews prior to interim and final submissions of designs and PS&E.
- Implement a communications plan between TxDOT, consultants, and contractors throughout the project life cycle.
- Keep the consultant contract active through construction and pre-construction meetings.
- Develop corrective action and recovery mechanisms to resolve E&O issues.

²² [TxDOT, Consultant Errors & Omissions Correction and Collection – Policy and Procedures \(2014\)](#)

- Establish consistency among organizational units while retaining independent decision-making.
- Utilize specific change order codes to track the source of E&O-related change orders.
- Develop training on how to address change orders associated with consultant E&O. TxDOT routinely invites members of the consultant community to participate in the class to provide a consultant perspective.
- Engage with the consultant community through workshops to discuss the process and answer questions.
- Create an internal working group between the department's PEPS, Construction, and Finance Divisions to support the districts in evaluating change orders, identifying the responsible party for resolving the E&O, and recovering additional costs caused by the E&O when appropriate.

TxDOT partnered with the Texas A&M Transportation Institute to create dedicated training that brings agency staff and the consultant community together to better understand the practice of managing E&O by working through real-world examples.

Major Considerations

The policy guiding TxDOT's E&O process was implemented as a result of a Texas government code passed to track and reduce costs related to E&O in engineering contracts.²³ While not a requirement to implementing processes, legal authority can be leveraged to gain support. Agencies similarly looking to establish processes to address E&O should consider existing legal authorities in their state or municipality. However, these may not always be necessary or may not be the more efficient option to reduce negative outcomes related to E&O.

Organizational structure, program size, typical project types, and local or regional industry are other important elements to consider when establishing processes intended to address E&O in project documents. Industry and consultant communities are important partners on transportation projects, and TxDOT's policy contains necessary provisions that provide for timely notification and coordination with the project consultants and contractors to create a culture of accountability. Strong relationships and industry feedback are essential to successfully developing and implementing policy and processes.

Results

TxDOT is tracking the outcomes of its E&O process to determine the extent of its impacts and realizing benefits from its collaborative resolution process and cooperative training efforts. By

²³ Texas Government Code Section 2252.905

HAZMAT USAGE LOG - FOR CONTRACTORS

All HAZMAT containing VOCs or HAPs shall be reported using this HAZMAT Usage Log to comply with Division Newport's Air Emissions Cap Permit, No. 1292 and associated regulations.

This log is due monthly, no later than the 10th day of the following month, or promptly at the completion of the project, whichever comes first.

I certify that: (1) All HAZMAT Safety data sheets have been reviewed and approved for use by NUWC Division Newport's Environmental and Safety Branches prior to bringing HAZMAT on the installation. (2) All architectural and industrial coatings (including, but not limited to: paints, primers, enamels, and sealants) shall comply with RIDEM Air Pollution Control Regulation (APCR) No. 33 and the limits listed at §33.7(B). (3) All adhesives must comply with RIDEM's APCR No. 44 and the emission limits listed at §44.7.1 and §44.7.2. (4) Solvents associated with the application or removal of adhesives and sealants must comply with the VOC and Vapor Pressure limits in, APCR No. §44.7(C) and §44.7(D).

SECTION 1: Information

1. Name:	2. Company:	3. Phone Number:
4. Email:		

SECTION 2: Table

	1a. DATE	1b. HAZMAT Name	1c. Product Purpose	1d. Manufacturer Name	1e. VOC Content (lbs./gal)	Gallons Used
X						
X						
X						
X						
X						
X						
X						
X						
X						

ADD

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF SHAREHOLDERS' EQUITY

(in thousands of dollars, except per share data)	Comm Shares	Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income (Loss)	Treasury Shares	Total
Balance January 1, 2000	\$4,928	\$104,891	\$483,463	\$ (43,524)	\$ (98,243)	\$451,515
Comprehensive income						
Net income			78,092			78,092
Minimum pension liability adjustment				(429)		(429)
Currency translation adjustment				(16,035)		(16,035)
Total comprehensive income						61,628
Cash dividends declared — \$0.57 per share			(24,157)			(24,157)
Net shares issued under certain benefit plans		2	(127)		645	520
Purchase of shares for treasury					(42,205)	(42,205)
Balance December 31, 2000	4,928	104,893	537,271	(59,988)	(139,803)	447,301
Comprehensive income						
Net income			83,589			83,589
Minimum pension liability adjustment				(514)		(514)
Unrealized gain on derivatives designated and qualified as cash flow hedges, net of tax				226		226
Currency translation adjustment				(6,450)		(6,450)
Total comprehensive income						76,851
Cash dividends declared — \$0.60 per share			(25,422)			(25,422)
Net shares issued under certain benefit plans		487	(737)		2,643	2,393
Purchase of shares for treasury					(2,425)	(2,425)
Balance December 31, 2001	4,928	105,380	594,701	(66,726)	(139,585)	498,698
Comprehensive income						
Net income			29,275			29,275
Minimum pension liability adjustment, net of tax of \$48,206				(79,697)		(79,697)
Unrealized loss on derivatives designated and qualified as cash flow hedges, net of tax				(246)		(246)
Currency translation adjustment				14,319		14,319
Total comprehensive income (loss)						(36,349)
Cash dividends declared — \$0.61 per share			(25,769)			(25,769)
Net shares issued under certain benefit plans		857	(712)		4,018	4,163
Purchase of shares for treasury					(11,590)	(11,590)
Balance December 31, 2002	\$4,928	\$106,237	\$597,495	\$(132,350)	\$(147,157)	\$429,153

See notes to these consolidated financial statements.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
CONSOLIDATED STATEMENTS OF CASH FLOWS

	Year Ended December 31		
	2002	2001	2000
	(In thousands of dollars)		
OPERATING ACTIVITIES			
Net income	\$ 29,275	\$ 83,589	\$ 78,092
Adjustments to reconcile net income to net cash provided by operating activities:			
Cumulative effect of a change in accounting principle, net of tax	37,607	—	—
Rationalization charges	10,468	—	—
Depreciation and amortization	37,040	36,205	34,712
Deferred income taxes	9,021	16,527	(2,286)
Loss (gain) on sale of fixed assets	476	(2,574)	(520)
Changes in operating assets and liabilities net of effects from acquisitions:			
Decrease (increase) in accounts receivable	8,786	(742)	1,905
Decrease in inventories	6,636	18,653	6,005
Decrease (increase) in other current assets	5,452	2,381	(4,659)
(Decrease) increase in accounts payable	(8,126)	4,647	(4,963)
(Decrease) increase in other current liabilities	(2,032)	(25,495)	12,030
(Decrease) in prepaid pension expense	(21,400)	(12,079)	(7,223)
Gross change in other long-term assets and liabilities	(2,005)	(775)	(305)
Other, net	(7,600)	(136)	8,046
NET CASH PROVIDED BY OPERATING ACTIVITIES	103,598	120,201	120,834
INVESTING ACTIVITIES			
Capital expenditures	(27,909)	(36,723)	(34,800)
Acquisitions of businesses and equity investments	(8,010)	(3,961)	(18,903)
Proceeds from sale of fixed assets	2,052	4,420	1,627
Purchase of marketable securities	(5,015)	(10)	—
Other	391	316	6
NET CASH (USED) BY INVESTING ACTIVITIES	(38,491)	(35,958)	(52,070)
FINANCING ACTIVITIES			
Proceeds from short-term borrowings	3,688	50,542	47,046
Payments on short-term borrowings	(10,926)	(44,299)	(48,972)
Notes payable to banks — net	(7,114)	(35,280)	29,270
Proceeds from long-term borrowings	150,172	71	54,294
Payments on long-term borrowings	(13,661)	(17,410)	(80,266)
Issuance of shares from treasury	3,294	1,891	645
Purchase of shares for treasury	(11,590)	(2,427)	(42,205)
Cash dividends paid	(25,390)	(25,418)	(24,034)
Other	—	(15)	(442)
NET CASH PROVIDED (USED) BY FINANCING ACTIVITIES	88,473	(72,345)	(64,664)
Effect of exchange rate changes on cash and cash equivalents	(997)	276	(1,456)
INCREASE IN CASH AND CASH EQUIVALENTS	152,583	12,174	2,644
Cash and cash equivalents at beginning of year	23,493	11,319	8,675
CASH AND CASH EQUIVALENTS AT END OF YEAR	\$176,076	\$ 23,493	\$ 11,319

See notes to these consolidated financial statements.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS
(In thousands of dollars except share and per share data)
December 31, 2002

NOTE A — SIGNIFICANT ACCOUNTING POLICIES

Principles of Consolidation: The consolidated financial statements include the accounts of Lincoln Electric Holdings, Inc., its wholly-owned and majority-owned subsidiaries and all non-majority owned entities for which it has a controlling interest (the “Company”) after elimination of all significant intercompany accounts, transactions and profits. Minority ownership interest in consolidated subsidiaries, which is not material, is recorded in Other long-term liabilities.

Cash Equivalents: The Company considers all highly liquid investments with a maturity of three months or less when purchased to be cash equivalents.

Inventories: Inventories are valued at the lower of cost or market. For domestic inventories, cost is determined principally by the last-in, first-out (LIFO) method, and for non-U.S. inventories, cost is determined by the first-in, first-out (FIFO) method. At December 31, 2002 and 2001, approximately 53% and 57%, respectively, of total inventories were valued using the LIFO method. The excess of current cost over LIFO cost amounted to \$39,924 at December 31, 2002 and \$39,703 at December 31, 2001.

Equity Investments: Investments in businesses in which the Company does not have a controlling interest and holds between a 20% and 50% ownership interest are accounted for using the equity method of accounting. Under the equity method, the investment is carried at cost plus the Company’s proportionate share of the net income or loss of the business since the date of acquisition.

Property, Plant and Equipment: Property, plant and equipment are stated at cost and include improvements which significantly extend the useful lives of existing plant and equipment. Depreciation and amortization are computed by both accelerated and straight-line methods over useful lives ranging from 3 to 20 years for machinery, tools and equipment, and up to 50 years for buildings. Net gains or losses related to asset dispositions are recognized in earnings in the period in which dispositions occur.

Goodwill and Intangibles: Prior to January 1, 2002, the Company amortized goodwill on a straight line basis over periods not exceeding 40 years. Goodwill had previously been tested for impairment under the provisions of Statement of Financial Accounting Standards (SFAS) No. 121, “*Accounting for the Impairment of Long-lived Assets and Long-lived Assets to be Disposed of*”. The Company previously evaluated goodwill for impairment by comparing the unamortized balance of goodwill to projected undiscounted cash flows, which did not indicate an impairment. Effective January 1, 2002, the Company adopted SFAS No. 142, “ *Goodwill and Other Intangible Assets*”. SFAS No. 142 requires cessation of goodwill amortization and a fair value approach to testing the impairment of goodwill and other intangibles. The Company has performed the required impairment tests of goodwill and indefinite-lived intangible assets as of January 1, 2002. As a result of these impairment tests, including third-party valuations based on acquisition prices of comparable businesses, a non-cash goodwill impairment charge of \$37,607 or \$0.88 per diluted share, net of tax benefits of \$700, has been recorded as a cumulative effect of a change in accounting principle. This non-cash charge was primarily due to weak market conditions in the European region and the Company’s expectations of lower financial performance at these reporting units. The Company performed its annual impairment test on October 1, 2002 and determined there was no additional impairment of the remaining goodwill.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

The following table reflects net income adjusted to exclude amortization of goodwill and the cumulative effect of a change in accounting principle, net of tax, in the periods presented:

(\$000's, except for earnings per share amounts)	For the Year Ended December 31,		
	2002	2001	2000
Net income, as reported	\$29,275	\$83,589	\$78,092
Add: Goodwill amortization	—	1,648	1,623
Adjusted net income	29,275	85,237	79,715
Add: Cumulative effect of a change in accounting principle, net of tax	37,607	—	—
Adjusted income before the cumulative effect of a change in accounting principle	\$66,882	\$85,237	\$79,715

The following table reflects Earnings Per Share adjusted to exclude amortization of goodwill and the cumulative effect of a change in accounting principle, net of tax, in the periods presented:

	For the Year Ended December 31,		
	2002	2001	2000
<u>Basic Earnings Per Share:</u>			
Net income, as reported	\$0.69	\$1.97	\$1.83
Add: Goodwill amortization	—	0.04	0.04
Adjusted net income	0.69	2.01	1.87
Add: Cumulative effect of a change in accounting principle, net of tax	0.89	—	—
Adjusted income before the cumulative effect of a change in accounting principle	\$1.58	\$2.01	\$1.87
<u>Diluted Earnings Per Share:</u>			
Net income, as reported	\$0.68	\$1.96	\$1.83
Add: Goodwill amortization	—	0.04	0.04
Adjusted net income	0.68	2.00	1.87
Add: Cumulative effect of a change in accounting principle, net of tax	0.88	—	—
Adjusted income before the cumulative effect of a change in accounting principle	\$1.56	\$2.00	\$1.87

The changes in the carrying amount of net goodwill by segment for the year ended December 31, 2002 are as follows:

	United States	Europe	Other Countries	Consolidated Total
Balance as of January 1, 2002	\$ 22,688	\$ 17,027	\$ 701	\$ 40,416
Foreign exchange effect on prior balances		(102)	7	(95)
Goodwill acquired during the period	—	1,982	99	2,081
Impairment loss from adoption of SFAS No. 142	(22,688)	(14,812)	(807)	(38,307)
Balance as of December 31, 2002	\$ —	\$ 4,095	\$ —	\$ 4,095

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

Gross intangible assets as of December 31, 2002 and 2001 were \$21,992 and \$12,907, respectively, which included accumulated amortization of \$9,238 and \$7,060, respectively. Aggregate amortization expense was \$1,028, \$1,066 and \$960 for 2002, 2001 and 2000, respectively.

Estimated annual intangible amortization expense in each of the next five years ending December 31 will be approximately \$1,000.

Long-lived Assets: During the first quarter of 2002, the Company adopted SFAS No. 144, “ *Accounting for the Impairment or Disposal of Long-Lived Assets*”. The Company previously evaluated long-lived assets for impairment under SFAS No. 121. SFAS No. 144 retains the requirements of SFAS No. 121 whereby the carrying value of long-lived assets is reviewed if facts and circumstances indicate a potential impairment of carrying value may have occurred utilizing relevant cash flow and profitability information. Impairment losses are recorded when the undiscounted cash flows estimated to be generated by those assets are less than the assets’ carrying amounts. The adoption of this Statement did not have a material impact on the financial statements of the Company.

Product Warranties: The Company accrues for product warranty claims based on historical experience and the expected material and labor costs to provide warranty service. The changes in the carrying amount of Product Warranty reserves for the year ended December 31, 2002 is as follows:

Balance at beginning of period	Charged to costs and expenses	Deductions	Balance at end of period
\$6,254	\$3,121	\$(3,363)	\$6,012

Revenue Recognition: The Company recognizes revenue when product is shipped and title is transferred to the customer.

Distribution Costs: Distribution costs, including warehousing and freight related to product shipments, is included in Cost of goods sold.

Stock-Based Compensation: Currently, the Company applies the intrinsic value method permitted under SFAS No. 123, “ *Accounting for Stock-Based Compensation*,” as defined in Accounting Principles Board (“APB”) Opinion No. 25, “ *Accounting for Stock Issued to Employees*” and related interpretations, in accounting for the Company’s stock option plans. Accordingly, no compensation cost has been recognized in the current or past years.

SFAS No. 123 requires pro forma disclosure of the effect on net income and earnings per share when applying the fair value method of valuing stock-based compensation. The following table sets forth the pro forma disclosure of net income and earnings per share using the Black-Scholes option pricing model. For purposes of this pro forma disclosure, the estimated fair value of the options is amortized ratably over the vesting periods.

	2002		2001		2000	
	Pro Forma	Reported	Pro Forma	Reported	Pro Forma	Reported
Net income	\$26,254	\$29,275	\$81,167	\$83,589	\$76,413	\$78,092
Basic earnings per share	0.62	0.69	1.92	1.97	1.79	1.83
Diluted earnings per share	0.61	0.68	1.90	1.96	1.79	1.83

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

Effective January 1, 2003, the Company will adopt the fair value method of recording stock options contained in SFAS No. 123, which is considered the preferable accounting method for stock-based employee compensation. All future employee stock option grants beginning January 1, 2003 will be expensed over the stock option vesting period based on the fair value at the date the options are granted. The Company will elect to expense stock options using the prospective method prescribed in SFAS No. 148, “ *Accounting for Stock-Based Compensation — Transition and Disclosure*”. The prospective method requires expense to be recognized for new grants or modifications issued beginning in the year of adoption. No expense is recognized in any year for options issued prior to adoption. The Company expects the impact from the adoption of this Statement to be approximately \$0.01 per share for 2003. The Company estimates the impact per share would increase annually over the next several years and level off at approximately \$0.06 per share by 2006.

Translation of Foreign Currencies: Asset and liability accounts are translated into U.S. dollars using exchange rates in effect at the date of the consolidated balance sheet; revenue and expense accounts are translated at monthly exchange rates. Translation adjustments are reflected as a component of Shareholders’ Equity. For subsidiaries operating in highly inflationary economies, both historical and current exchange rates are used in translating balance sheet accounts, and translation adjustments are included in net income.

Transaction gains and losses are included in Selling, general & administrative expenses and were not material.

Financial Instruments: The Company, on a limited basis, uses forward exchange contracts to hedge exposure to exchange rate fluctuations on certain intercompany loans, purchase and sales transactions and other intercompany commitments. Contracts are written on a short-term basis and are not held for trading or speculative purposes.

The Company recognizes derivative instruments as either assets or liabilities in the balance sheets at fair value. The accounting for changes in the fair value of derivative instruments depends on whether it has been designated and qualifies as part of a hedging relationship and further, on the type of hedging relationship.

For derivative instruments that qualify as a fair value hedge (i.e., hedging the exposure to changes in the fair value of an asset or a liability), the gain or loss on the derivative instrument, as well as the offsetting loss or gain on the hedged item attributable to the hedged risk are recognized in earnings. For derivative instruments that qualify as a cash flow hedge (i.e., hedging the exposure to variability in expected future cash flows), the effective portion of the gain or loss on the derivative instrument is reported as a component of Other comprehensive income and reclassified into earnings in the same period or periods during which the hedged transaction affects earnings. Any remaining gain or loss on the derivative instrument is recognized in earnings. The Company does not hedge its net investments in foreign subsidiaries. For derivative instruments not designated as hedges, the gain or loss from changes in their fair values is recognized in earnings.

Research and Development: Research and development costs are expensed as incurred, and totaled \$19,150 in 2002, \$17,946 in 2001 and \$16,604 in 2000.

Estimates: The preparation of financial statements in conformity with accounting principles generally accepted in the United States, requires management to make estimates and assumptions in certain circumstances that affect the amounts reported in the accompanying consolidated financial statements and notes. Actual results could differ from these estimates.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

Earnings per Share: The following table sets forth the computation of basic and diluted earnings per share (dollars and shares in thousands, except per share amounts).

	2002	2001	2000
Numerator:			
Income before the cumulative effect of a change in accounting principle	\$ 66,882	\$83,589	\$78,092
Cumulative effect of a change in accounting principle, net of tax	(37,607)	—	—
Net income	\$ 29,275	\$83,589	\$78,092
Denominator:			
Denominator for basic earnings per share - Weighted-average shares outstanding	42,259	42,375	42,670
Effect of dilutive securities — Employee stock options	540	235	20
Denominator for diluted earnings per share - Adjusted weighted-average shares outstanding	42,799	42,610	42,690
Basic earnings per share			
Income before the cumulative effect of a change in accounting principle	\$ 1.58	\$ 1.97	\$ 1.83
Cumulative effect of a change in accounting principle, net of tax	(0.89)	—	—
Basic earnings per share	\$ 0.69	\$ 1.97	\$ 1.83
Diluted earnings per share			
Income before the cumulative effect of a change in accounting principle	\$ 1.56	\$ 1.96	\$ 1.83
Cumulative effect of a change in accounting principle, net of tax	(0.88)	—	—
Diluted earnings per share	\$ 0.68	\$ 1.96	\$ 1.83

Reclassification: Certain reclassifications have been made to prior-year financial statements to conform to current year classifications.

New Accounting Pronouncements: Effective January 1, 2002, the Company adopted SFAS No. 144, “ *Accounting for the Impairment or Disposal of Long-Lived Assets*”. SFAS No. 144 supercedes SFAS No. 121, and APB Opinion No. 30, “ *Reporting the Results of Operations — Reporting the Effects of Disposal of a Segment of a Business, and Extraordinary, Unusual and Infrequently Occurring Events*”. SFAS No. 144 retains the requirements of SFAS No. 121 whereby an impairment loss is recognized if the carrying value of the asset is not recoverable from its undiscounted cash flows or fair values are less than carrying values. SFAS No. 144 broadens the scope of APB Opinion No. 30 provisions for the presentation of discontinued operations to include a component of an entity. The Statement requires a component of an entity sold or considered held for sale be presented as a discontinued operation. In addition, expected future operating losses from discontinued operations must be reflected in the periods incurred, rather than at the measurement date as previously required by APB Opinion No. 30. The adoption of this Statement did not have a material impact on the financial statements of the Company.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

In April 2002, the FASB issued SFAS No. 145, “*Rescission of FASB Statements No. 4, 44 and 64, Amendment of FASB Statement No. 13, and Technical Corrections*”. SFAS No. 145 is effective for fiscal years beginning after May 15, 2002. SFAS No. 145 requires that debt extinguishment must meet the criteria under APB Opinion No. 30 to be classified as an extraordinary item. This statement also amends SFAS No. 13 to require sale-leaseback accounting for certain lease modifications that have economic effects similar to sale-leaseback transactions. The Company has not yet determined the impact, if any, this Statement will have on the financial statements of the Company.

In June 2002, the FASB issued SFAS No. 146, “*Accounting for Exit or Disposal Activities*”. SFAS No. 146 is effective for disposal activities initiated after December 31, 2002. SFAS No. 146 requires liabilities for one-time termination benefits incurred over future service periods be measured at fair value as of the termination date and recognized over the future service period. This Statement also requires liabilities associated with disposal activities be recorded when incurred instead of when probable as currently required by SFAS No. 5 “*Accounting for Contingencies*”. These liabilities should be adjusted for subsequent changes resulting from revisions to either the timing or amount of estimated cash flows, discounted at the original credit-adjusted risk-free rate. Interest on the liability would be accreted and charged to expense as an operating item. The Company has not yet determined the impact, if any, this Statement will have on the financial statements of the Company.

In November 2002, the FASB issued Interpretation No. 45, “*Guarantor’s Accounting and Disclosure Requirements for Guarantees, Including Indirect Guarantees of Indebtedness of Others*”. Interpretation No. 45 requires certain disclosures to be made by a guarantor in its financial statements about its obligations under certain guarantees, including product warranties. The disclosure provisions of this Statement are effective for the Company as of the fourth quarter of 2002. The Company has made the required disclosures in these financial statements. This Statement also requires a guarantor to recognize, at inception, for all guarantees issued or modified after December 31, 2002, a liability for the fair value of the obligations it has undertaken in issuing a guarantee. The adoption of the fair value provisions of this Statement are not expected to have a material impact on the financial statements of the Company.

In December 2002, the FASB issued SFAS No. 148, “*Accounting for Stock-Based Compensation — Transition and Disclosure*”. SFAS No. 148 does not alter the provision of SFAS No. 123, nor does it require stock-based compensation to be measured under the fair-value method. Rather, SFAS No. 148 does provide three alternative transition methods to companies that elect to expense stock-based compensation using the fair-value approach under SFAS No. 123. As mentioned previously, the Company will adopt the fair-value method of recording stock options contained in SFAS No. 123 effective January 1, 2003. The Company will elect to expense stock options using the prospective method prescribed in SFAS No. 148. The prospective method requires expense to be recognized for new grants or modifications issued beginning in the year of adoption. No expense is recognized in any year for options issued prior to adoption. The Company expects the impact from the adoption of this Statement to be approximately \$0.01 per share for 2003. The Company estimates the impact per share would increase annually over the next several years and level off at approximately \$0.06 per share by 2006.

In December 2002, the Emerging Issues Task Force (“EITF”) issued EITF Issue 00-21, “*Accounting for Revenue Arrangements with Multiple Deliverables*”. EITF 00-21 provides guidance on determining whether a revenue arrangement contains multiple deliverable items and if so, requires revenue be allocated amongst the different items based on fair value. EITF 00-21 also requires revenue on any item in a revenue arrangement with multiple deliverables not delivered completely must be deferred until delivery of the item is completed. The effective date of this Issue for the Company will be July 1, 2003. The Company has not yet determined the impact, if any, this Statement will have on the financial statements of the Company.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE A — SIGNIFICANT ACCOUNTING POLICIES (Continued)

In January 2003, the FASB issued Interpretation No. 46, “ *Consolidation of Variable Interest Entities*”. Interpretation No. 46 provides guidance for identifying a controlling interest in a Variable Interest Entity (“VIE”) established by means other than voting interests. Interpretation No. 46 also requires consolidation of a VIE by an enterprise that holds such a controlling interest. The effective date for this Interpretation will be July 1, 2003. The Company has not yet determined the impact, if any, this Interpretation will have on the financial statements of the Company.

Other: Included in Selling, general & administrative expenses are the costs related to the Company’s discretionary employee bonus, net of hospitalization costs, of \$32,218 in 2002, \$31,947 in 2001 and \$54,509 in 2000.

NOTE B — SHAREHOLDERS’ EQUITY

In 1999, the Board of Directors authorized an additional share repurchase program of up to 5,000,000 shares of the Company’s outstanding Common Shares to satisfy obligations under its stock option plans. This share repurchase program is in addition to the 5,000,000 shares authorized for repurchase by the Board of Directors in September 1998. The Company purchased 501,330 shares at an average cost of \$23.12 per share in 2002, bringing the total shares purchased through December 31, 2002, to 7,740,789 at an average cost of \$20.30 per share.

NOTE C — ACCUMULATED OTHER COMPREHENSIVE INCOME (LOSS)

The components of accumulated other comprehensive income (loss) are as follows:

	Minimum Pension Liability Adjustment, net of tax	Currency Translation Adjustment	Unrealized Gain (Loss) on Derivatives Designated and Qualified as Cash Flow Hedges, net of tax	Total Accumulated Other Comprehensive Loss
Balance January 1, 2000	\$ (792)	\$(42,732)	\$ —	\$ (43,524)
Other comprehensive (loss)	(429)	(16,035)	—	(16,464)
Balance December 31, 2000	(1,221)	(58,767)	—	(59,988)
Other comprehensive (loss) income	(514)	(6,450)	226	(6,738)
Balance December 31, 2001	(1,735)	(65,217)	226	(66,726)
Other comprehensive (loss) income	(79,697)	14,319	(246)	(65,624)
Balance December 31, 2002	\$(81,432)	\$(50,898)	\$ (20)	\$(132,350)

NOTE D — STOCK PLANS

The 1998 Stock Option Plan (“Stock Option Plan”) was adopted by the shareholders to replace The Lincoln Electric Company 1988 Incentive Equity Plan (“Incentive Equity Plan”) which expired in May 1998. The Stock Option Plan provides for the grant of options for 5,000,000 shares of Company stock to key employees over a ten-year period.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE D — STOCK PLANS (Continued)

The following table summarizes the option activity for the three years ended December 31, 2002, under all Plans:

	2002		2001		2000	
	Options	Weighted-Average Exercise Price	Options	Weighted-Average Exercise Price	Options	Weighted-Average Exercise Price
Balance, beginning of year	2,736,251	\$18.12	2,218,793	\$16.76	1,567,334	\$18.13
Options granted	669,900	\$22.97	694,390	\$21.65	787,375	\$13.60
Options exercised	(222,246)	\$15.21	(171,082)	\$14.91	(29,876)	\$14.08
Options canceled	(4,434)	\$19.57	(5,850)	\$17.56	(106,040)	\$14.36
Balance, end of year	3,179,471	\$19.34	2,736,251	\$18.12	2,218,793	\$16.76
Exercisable at end of year	1,875,464	\$18.29	1,430,280	\$17.75	1,065,512	\$17.82

During 1996, options for 335,180 shares were granted to employees in settlement of a lawsuit over performance awards relating to prior years. Exercise prices are \$15.00 and \$17.00 per share. These options are exercisable over five- and ten-year periods and are fully vested, non-qualified and non-transferable. At December 31, 2002 and 2001, there were 94,756 and 124,554, respectively, of these options outstanding.

All other options granted under both the Stock Option Plan and the Incentive Equity Plan are outstanding for a term of ten years from the date of grant. The majority of the options granted under both plans vest ratably over a period of three years from the grant date. The exercise prices of all options were equal to the fair market value of the Company’s shares at the date of grant. As permitted under SFAS No. 123, the Company has continued to record stock-based compensation in accordance with the intrinsic value method established by APB Opinion No. 25. Under the intrinsic value method, compensation expense is measured as the excess, if any, of the market price at the date of grant over the exercise price of the options. Accordingly, no compensation expense was recognized upon the award of these stock options.

In estimating the fair value of options granted during 2002 for the Stock Option Plan and the Incentive Equity Plan, an expected option life of five years was used based on the Company’s historical experience. In prior years, the Company had used ten years as the expected option life. The other weighted-average assumptions were as follows:

	2002	2001	2000
Expected volatility	43.50%	43.13%	42.60%
Dividend yield	2.76%	2.45%	2.90%
Risk-free interest rate	3.60%	5.11%	5.17%
Weighted-average fair value of options granted during the year	\$ 7.63	\$ 9.52	\$ 5.59

The Stock Option Plan for Non-Employee Directors (“Directors Stock Option Plan”) was adopted in May 2000 to replace The Lincoln Non-Employee Directors Restricted Stock Plan, which was terminated. The Directors Stock Option Plan provides for the grant of stock options for the purchase of up to an aggregate of 500,000 Common Shares. Options issued under the Directors Stock Option Plan were 30,000 in 2002 and 34,000 in 2001. There were no shares issued in connection with The Lincoln Non-Employee Directors Restricted Stock Plan during 2002 or 2001 and 5,335 in 2000. Forfeitures under the service requirements of the plan were 1,644 in 2000.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE D — STOCK PLANS (Continued)

At December 31, 2002, there were 2,594,535 shares of common stock available for future grant under all plans, and the weighted-average remaining contractual life of outstanding options was 7.6 years. The following table summarizes information about stock options outstanding as of December 31, 2002:

Exercise Price Range	Outstanding		Exercisable		Weighted-Average Remaining Life
	Number of Options	Weighted-Average Exercise Price	Number of Options	Weighted-Average Exercise Price	
\$13 - \$15	691,148	\$13.51	456,397	\$13.52	7.4
\$15 - \$17	164,184	\$15.00	164,184	\$15.00	3.6
\$17 - \$19	170,206	\$17.40	160,706	\$17.39	4.6
\$19 - \$21	521,542	\$19.75	511,708	\$19.77	6.5
\$21 - \$22	849,982	\$21.58	263,435	\$21.61	9.0
\$22 - \$24	758,409	\$23.03	300,034	\$22.41	8.4
\$24 - \$26	24,000	\$25.50	—	—	9.9

The 1995 Lincoln Stock Purchase Plan provides employees the ability to purchase open market shares on a commission-free basis up to a limit of ten thousand dollars annually. Under this plan, 400,000 shares have been authorized to be purchased. There were 53,035, 6,271 and 26,559 shares purchased in 2002, 2001 and 2000, respectively.

NOTE E — RATIONALIZATION CHARGES

During the first quarter of 2002, the Company recorded rationalization charges of \$10,468 (\$7,045 after-tax), or \$0.17 per diluted share. The rationalization charges were principally related to a voluntary retirement program affecting approximately 3% of the Company’s U.S. workforce and asset impairment charges. Workforce reduction charges were \$5,353, while non-cash asset impairment charges were \$5,115.

NOTE F — SHORT-TERM AND LONG-TERM DEBT

At December 31, 2002 and 2001, long-term debt consisted of the following:

	2002	2001
Senior Unsecured Notes due 2007, interest at 5.58%	\$ 40,000	\$ —
Senior Unsecured Notes due 2009, interest at 5.89%	30,000	—
Senior Unsecured Notes due 2012, interest at 6.36%	80,000	—
8.73% Senior Note due 2003	9,375	18,750
Foreign borrowings due through 2011, interest at 2.3% to 10.0% (2.6% to 12.4% in 2001)	8,409	10,725
Capital leases	5,991	3,891
Other borrowings due through 2023, interest at 2.0% to 7.9%	13,804	4,249
	187,579	37,615
Less current portion	13,433	13,434
Total	\$174,146	\$24,181

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE F — SHORT-TERM AND LONG-TERM DEBT (Continued)

During March 2002, the Company issued Senior Unsecured Notes (the “Notes”) totaling \$150,000 through a private placement. The Notes have maturities ranging from five to ten years with a weighted average interest rate of 6.1% and an average tenure of eight years. Interest is payable semi-annually in March and September. The proceeds will be used for general corporate purposes. The Notes contain certain affirmative and negative covenants, including restrictions on asset dispositions and financial covenants (interest coverage and funded debt-to-Earnings Before Interest, Taxes, Depreciation and Amortization (“EBITDA”) ratios). The Notes will not be and have not been registered under the Securities Act of 1933, as amended, and may not be offered or sold in the United States absent registration or an applicable exemption from registration requirements.

During March 2002, the Company entered into floating rate interest rate swap agreements with notional amounts totaling \$80 million, to convert a portion of the outstanding Notes from fixed to floating rates based on six-month London Inter-bank Offered Rate (“LIBOR”), plus a spread of between 15.5 and 37.5 basis points. The variable rates are reset every six months, at which time payment or receipt of interest is settled. These swaps were designated as fair value hedges, and as such, the gain or loss on the derivative instrument, as well as the offsetting gain or loss on the hedged item attributable to the hedged risk are recognized in earnings. Net payments or receipts under these agreements are recognized as adjustments to interest expense. The fair value of these swaps at December 31, 2002 was \$9,893 and is included in Other assets.

During April 2002, the Company replaced its prior committed revolving credit facility with a new three-year revolving Credit Agreement that provides up to \$125,000 in borrowings and expires in April 2005. This Credit Agreement may be used for general corporate purposes. The interest rate on borrowings under the Credit Agreement is based on either the LIBOR plus a spread based on the Company’s leverage ratio or the prime rate, at the Company’s election. A facility fee is payable based upon the daily aggregate amount of commitments. The facility fee is based on the Company’s leverage ratio. The Credit Agreement provides for the issuance of Letters of Credit, subject to limits based on amounts outstanding under the Credit Agreement and is subject to the same covenants as the Notes. As of December 31, 2002, there are no borrowings under the credit agreement.

During April 2002, the Company amended and restated its existing 8.73% Senior Notes due in 2003. The 8.73% Senior Notes were amended and restated so that the affirmative and negative covenants are consistent with the Notes. In addition, the amendment and restatement of the 8.73% Senior Notes added an uncommitted private shelf facility allowing for the issuance of an aggregate of \$100,000 of additional senior unsecured notes (the “Shelf Notes”). There were no Shelf Notes issued or outstanding under the private shelf facility as of December 31, 2002.

In addition to committed facilities discussed above, at December 31, 2002, the Company had short-term credit lines in the United States, with uncommitted borrowing capacity of \$20,000. Short-term borrowings of foreign subsidiaries, included in Notes payable to banks, were \$945 and \$13,163 at December 31, 2002 and 2001, at weighted-average interest rates of 9.7% and 8.1%, respectively. At December 31, 2002 and 2001, \$13,083 and \$7,704 of indebtedness was secured by property, plant and equipment.

Maturities of long-term debt, including payments under capital leases, for the five years succeeding December 31, 2002 are \$13,433 in 2003, \$3,822 in 2004, \$1,817 in 2005, \$1,825 in 2006, \$46,119 in 2007 and \$120,563 thereafter. Total interest paid was \$8,343 in 2002, \$5,447 in 2001 and \$6,957 in 2000.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE G — INCOME TAXES

The components of income before income taxes are as follows:

	2002	2001	2000
U.S	\$64,049	\$ 96,397	\$105,181
Non-U.S	23,894	19,644	16,496
Total	\$87,943	\$116,041	\$121,677

Components of income tax expense (benefit) are as follows:

	2002	2001	2000
Current:			
Federal	\$10,687	\$10,029	\$32,159
Non-U.S	4,116	5,266	9,278
State and local	1,271	614	4,198
	16,074	15,909	45,635
Deferred:			
Federal	2,733	12,700	333
Non-U.S	2,511	1,604	(2,051)
State and local	(257)	2,239	(332)
	4,987	16,543	(2,050)
Total	\$21,061	\$32,452	\$43,585

The differences between total income tax expense and the amount computed by applying the statutory Federal income tax rate to income before income taxes were as follows:

	2002	2001	2000
Statutory rate of 35% applied to pre-tax income	\$30,780	\$40,614	\$42,587
Effect of state and local income taxes, net of Federal tax benefit	569	1,854	2,513
Taxes (less than) in excess of the U.S. tax rate on non-U.S. earnings, including utilization of tax loss carryforwards and losses with no benefit	(1,736)	(5)	1,454
Extraterritorial income exclusion/foreign sales corporation	(1,347)	(5,319)	(1,437)
U.S. tax (benefit) cost of foreign source income	(6,623)	2,136	(95)
Other — net	(582)	(6,828)	(1,437)
Total	\$21,061	\$32,452	\$43,585
Effective tax rate	23.9%	28.0%	35.8%

Total income tax payments, net of refunds, were \$9,818 in 2002, \$25,404 in 2001 and \$35,699 in 2000.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE G — INCOME TAXES (Continued)

Significant components of deferred tax assets and liabilities at December 31, 2002 and 2001, were as follows:

	2002	2001
Deferred tax assets:		
Tax loss and credit carryforwards	\$ 19,453	\$ 18,245
State income taxes	3,158	2,541
Other accruals	9,075	14,948
Employee benefits	5,753	6,380
Pension obligations	35,766	5,183
Other	20,233	13,884
	93,438	61,181
Valuation allowance	(14,993)	(14,198)
	78,445	46,983
Deferred tax liabilities:		
Property, plant and equipment	(39,096)	(35,338)
Pension obligations	(1,141)	(14,945)
Other	(16,065)	(16,454)
	(56,302)	(66,737)
Total	\$ 22,143	\$(19,754)

At December 31, 2002, certain non-U.S. subsidiaries had tax loss carryforwards of approximately \$53,017 that will expire in various years from 2003 through 2011, except for \$28,591 for which there is no expiration date.

The Company does not provide deferred income taxes on unremitted earnings of non-U.S. subsidiaries, as such funds are deemed permanently reinvested. It is not practicable to calculate the deferred taxes associated with the remittance of these investments.

In assessing the realizability of deferred tax assets, the Company assesses whether it is more likely than not that a portion or all of the deferred tax assets will not be realized. The Company considers the scheduled reversal of deferred tax liabilities, projected future taxable income, and tax planning strategies in making this assessment. At December 31, 2002 a valuation allowance of \$14,993 relating principally to foreign tax loss and credit carryforwards, has been recorded against these deferred tax assets based on this assessment. The Company believes it is more likely than not that the tax benefit of the net deferred tax assets will be realized. The amount of net deferred tax assets considered realizable could be increased or reduced in the future if the Company’s assessment of future taxable income or tax planning strategies changes.

NOTE H — RETIREMENT ANNUITY AND GUARANTEED CONTINUOUS EMPLOYMENT PLANS

The Company and its subsidiaries maintain a number of defined benefit and defined contribution plans to provide retirement benefits for employees in the U.S., as well as employees outside the U.S. These plans are maintained and contributions are made in accordance with the Employee Retirement Income Security Act of 1974, local statutory law or as determined by the Board of Directors. The plans generally provide benefits based upon years of service and compensation. Pension plans are funded except for a domestic non-qualified pension plan for certain key employees. Substantially all U.S. employees are covered under a 401(k) savings plan in which they may invest 1% or more of eligible compensation, limited to maximum amounts as determined by the Internal Revenue Service. The plan provides for Company matching contributions of 35% of the first 6% of employee compensation contributed to

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE H — RETIREMENT ANNUITY AND GUARANTEED CONTINUOUS EMPLOYMENT PLANS (Continued)

the plan. The plan includes a feature in which participants could elect to receive an annual Company contribution of 2% of their base pay in exchange for forfeiting certain benefits under the pension plan.

The changes in the pension plans’ benefit obligations were as follows:

	2002	2001
Obligation at January 1	\$475,418	\$445,634
Service cost	15,018	11,768
Interest cost	33,264	31,783
Participant contributions	773	667
Plan amendments	577	189
Actuarial loss	26,915	10,062
Benefit payments	(25,447)	(23,148)
Currency translation	5,181	(1,537)
Obligation at December 31	\$531,699	\$475,418

The changes in the fair values of the pension plans’ assets were as follows:

	2002	2001
Plan assets at January 1	\$428,460	\$452,697
Actual return on plan assets	(45,893)	(12,057)
Employer contributions	21,400	12,079
Participant contributions	786	668
Benefit payments	(25,447)	(23,148)
Currency translation	4,774	(1,779)
Plan assets at December 31	\$384,080	\$428,460

The funded status of the pension plans was as follows:

	2002	2001
Plan assets (less than) projected benefit obligations	\$(147,619)	\$(46,958)
Unrecognized net loss	178,682	67,594
Unrecognized prior service cost	7,711	8,531
Unrecognized transition assets, net	(476)	(861)
Net amount recognized in the balance sheets	\$ 38,298	\$ 28,306

The amounts recognized in the consolidated balance sheets was composed of:

	2002	2001
Prepaid pension cost	\$ 3,345	\$ 38,784
Accrued pension liability	(102,881)	(13,779)
Intangible asset	7,620	1,566
Accumulated other comprehensive income	130,214	1,735
Net amount recognized in the balance sheets	\$ 38,298	\$ 28,306

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE H — RETIREMENT ANNUITY AND GUARANTEED CONTINUOUS EMPLOYMENT PLANS (Continued)

A minimum pension liability adjustment was recorded during the fourth quarter as a liability with a corresponding decrease to shareholders' equity. At December 31, 2002, the Company recorded an after-tax charge to shareholders' equity of \$79,697.

The projected benefit obligation, accumulated benefit obligation, and fair value of plan assets for the U.S. pension plans with accumulated benefit obligations in excess of plan assets were \$482,943, \$439,604 and \$346,207, respectively; as of December 31, 2002 and \$13,100, \$12,863 and \$0, respectively as of December 31, 2001. The projected benefit obligation, accumulated benefit obligation, and fair value of plan assets for the non-U.S. pension plans with accumulated benefit obligations in excess of plan assets were \$18,455, \$17,186 and \$11,874, respectively; as of December 31, 2002 and \$15,994, \$14,842 and \$11,953, respectively as of December 31, 2001.

A summary of the components of total pension expense was as follows:

	2002	2001	2000
Service cost — benefits earned during the year	\$ 13,393	\$ 11,768	\$ 12,321
Interest cost on projected benefit obligation	33,264	31,783	31,077
Expected return on plan assets	(37,461)	(40,117)	(40,733)
Amortization of transition asset	(437)	(426)	(436)
Amortization of prior service cost	1,377	1,339	1,177
Amortization of net loss (gain)	340	(21)	(209)
Net pension cost of defined benefit plans	10,476	4,326	3,197
Defined contribution plans	4,387	3,629	2,040
Total pension expense	\$ 14,863	\$ 7,955	\$ 5,237

Weighted-average assumptions used in accounting for the defined benefit plans as of December 31, 2002 and 2001, were as follows:

	2002	2001
Discount rate	6.7%	7.2%
Rate of increase in compensation	4.9%	4.9%
Expected return on plan assets	9.1%	9.1%

The Company anticipates lowering the expected rate of return on plan assets to approximately 8.8% for 2003. U.S. plan assets consist of fixed income and equity securities. Non-U.S. plan assets are invested in non-U.S. insurance contracts and non-U.S. equity and fixed income securities. The Company does not have, and does not provide for, any postretirement or postemployment benefits other than pensions.

The Cleveland, Ohio, area operations have a Guaranteed Continuous Employment Plan covering substantially all employees which, in general, provides that the Company will provide work for at least 75% of every standard work week (presently 40 hours). This plan does not guarantee employment when the Company's ability to continue normal operations is seriously restricted by events beyond the control of the Company. The Company has reserved the right to terminate this plan effective at the end of a calendar year by giving notice of such termination not less than six months prior to the end of such year.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE I — SEGMENT INFORMATION

The Company’s primary business is the design, manufacture and sale, in the U.S. and international markets, of arc, cutting and other welding products. The Company manages its operations by geographic location and has three reportable segments: the United States, Europe and all Other Countries. Each reporting segment is managed separately because each faces a distinct economic environment, a different customer base and a varying level of competition and market sophistication. Segment performance and resource allocation is measured based on income before interest and income taxes. The accounting policies of the reportable segments are the same as those described in Note A — Significant Accounting Policies. Financial information for the reportable segments follows:

	United States	Europe	Other Countries	Eliminations	Consolidated
2002:					
Net sales to unaffiliated customers	\$612,425	\$202,373	\$179,279	\$ —	\$994,077
Inter-segment sales	54,291	16,457	30,892	(101,640)	—
Total	\$666,716	\$218,830	\$210,171	\$(101,640)	\$994,077
Income before interest and income taxes	\$ 61,449	\$ 14,271	\$ 17,344	\$ 696	\$ 93,760
Interest income					3,239
Interest expense					(9,056)
Income before income taxes					\$ 87,943
Total assets	\$598,779	\$204,266	\$174,542	\$ (76,318)	\$901,269
Capital expenditures	18,600	4,480	4,829	—	27,909
Depreciation and amortization	25,176	6,040	5,824	—	37,040
2001:					
Net sales to unaffiliated customers	\$639,914	\$180,087	\$158,876	\$ —	\$978,877
Inter-segment sales	57,015	9,893	29,447	(96,355)	—
Total	\$696,929	\$189,980	\$188,323	\$ (96,355)	\$978,877
Income before interest and income taxes	\$ 91,548	\$ 13,194	\$ 15,860	\$ (146)	\$120,456
Interest income					1,122
Interest expense					(5,537)
Income before income taxes					\$116,041
Total assets	\$488,992	\$180,101	\$164,413	\$ (52,195)	\$781,311
Capital expenditures	26,620	5,365	3,914	824	36,723
Depreciation and amortization	25,696	5,763	5,347	(601)	36,205

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE I — SEGMENT INFORMATION (Continued)

	United States	Europe	Other Countries	Eliminations	Consolidated
2000:					
Net sales to unaffiliated customers	\$705,086	\$185,340	\$168,175	\$ —	\$1,058,601
Inter-segment sales	70,146	12,108	21,936	(104,190)	—
Total	\$775,232	\$197,448	\$190,111	\$(104,190)	\$1,058,601
Income before interest and income taxes	\$104,238	\$ 12,458	\$ 10,650	\$ 982	\$ 128,328
Interest income					732
Interest expense					(7,383)
Income before income taxes					\$ 121,677
Total assets	\$507,826	\$184,703	\$189,253	\$ (91,503)	\$ 790,279
Capital expenditures	20,742	3,545	10,425	88	34,800
Depreciation and amortization	23,806	6,191	5,381	(666)	34,712

The United States segment for 2002 includes rationalization charges of \$8,358, while the European and Other Countries segments include rationalization charges of \$1,057 and \$1,053, respectively (see Note E). The European segment for 2001 includes a pre-tax gain of \$3,087 relating to the sale of property and a pre-tax charge of \$1,144 relating to severance and redundancy charges. The United States segment for 2000 included a net charge of \$13,399 for costs associated with the lapsed Charter plc offer, net of proceeds from settlement of a dispute with one of the Company’s product liability insurance carriers.

Inter-segment sales between reportable segments are accounted for at prices comparable to normal customer sales and are eliminated in consolidation. Export sales (excluding intercompany sales) from the United States were \$63,248 in 2002, \$58,489 in 2001 and \$60,223 in 2000. No individual customer comprised more than 10% of the Company’s total revenues for the three years ended December 31, 2002.

The geographic split of the Company’s net sales, based on country of origin, and property, plant and equipment was as follows:

	2002	2001	2000
Net sales:			
United States	\$549,178	\$581,425	\$ 644,863
Foreign countries	444,899	397,452	413,738
Total	\$994,077	\$978,877	\$1,058,601
Property, plant and equipment:			
United States	\$166,339	\$177,823	\$ 172,838
Foreign countries	108,780	97,434	99,706
Eliminations	(3,266)	(4,286)	(3,531)
Total	\$271,853	\$270,971	\$ 269,013

Net sales derived from customers and property, plant and equipment in any individual foreign country were not material for disclosure.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE J — ACQUISITIONS

During January 2002, the Company acquired 85% of Bester S.A., a leading manufacturer of welding equipment located in Poland, for \$7,780, including assumed debt. During 2002, the Company also acquired an additional 6% of Bester through additional investments totaling \$3,500, of which \$1,500 has been paid as of December 31, 2002.

In December 2001, the Company acquired 100% of Messer Soldaduras de Venezuela S.A., a leading manufacturer of consumable welding products located in Venezuela, for \$3,382, including assumed debt.

On April 26, 2000, the Company made a cash offer in the United Kingdom to purchase all of the outstanding shares of Charter plc, a British industrial holding company. In October 2000, the Company’s offer to purchase Charter plc lapsed. As a result, the acquisition was not completed and the Company recorded an additional after-tax charge of \$11,608 (\$0.27 per diluted share) during the fourth quarter of 2000 representing remaining costs associated with the lapsed bid. For the year, the Company recorded total non-recurring charges of \$13,399 (\$8,126 after-tax) in Other income (expense). Of this amount, the quarter ended June 30, 2000, included a net gain of \$10,183 (\$6,273 after-tax) principally related to proceeds received in settlement of a dispute with one of the Company’s product liability insurance carriers. In addition, the quarter ended September 30, 2000, included a net charge of \$4,396 (\$2,791 after-tax) principally related to costs of foreign currency options purchased in connection with the lapsed Charter plc bid. During the period in which the Charter acquisition was pending, the Company had suspended dividend payments and its share repurchase program; both were re-instituted in December 2000 upon lapse of the offer.

During the first quarter of 2000, the Company acquired a 35% equity interest in Kuang Tai, a Taiwan-based manufacturer of welding wire, for \$16,774 and 100% of C.I.F.E. S.r.l., an Italian-based manufacturer of MIG wire, for \$19,510, including assumed debt, which was accounted for as a purchase.

NOTE K — FAIR VALUES OF FINANCIAL INSTRUMENTS

The Company has various financial instruments, including cash, cash equivalents, short- and long-term debt and forward contracts. The Company has determined the estimated fair value of these financial instruments by using available market information and appropriate valuation methodologies that require judgment.

The Company enters into forward exchange contracts to hedge foreign currency transactions on a continuing basis for periods consistent with its exposures. This hedging minimizes the impact of foreign exchange rate movements on the Company’s operating results. The total notional value of forward currency exchange contracts was \$26,060 at December 31, 2002 and \$41,032 at December 31, 2001, which approximated fair value.

The Company entered into floating rate interest rate swap agreements with notional amounts totaling \$80,000, to convert a portion of the outstanding Notes from fixed to floating rates (see Note D). The fair value of these swaps at December 31, 2002 was \$9,893.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE K — FAIR VALUES OF FINANCIAL INSTRUMENTS (Continued)

The carrying amounts and estimated fair value of the Company’s significant financial instruments at December 31, 2002 and 2001, were as follows:

	December 31, 2002		December 31, 2001	
	Carrying Amounts	Fair Value	Carrying Amounts	Fair Value
Cash and cash equivalents	\$176,076	\$176,076	\$23,493	\$23,493
Marketable securities	4,881	4,881	52	52
Interest rate swaps	9,893	9,893	—	—
Notes payable to banks	945	945	13,163	13,163
Long-term debt (including current portion)	187,579	193,158	37,615	37,289

NOTE L — OPERATING LEASES

The Company leases sales offices, warehouses and distribution centers, office equipment and data processing equipment. Such leases, some of which are noncancelable and, in many cases, include renewals, expire at various dates. The Company pays most maintenance, insurance and taxes relating to leased assets. Rental expense was \$10,150 in 2002, \$8,887 in 2001 and \$8,931 in 2000.

At December 31, 2002, total future minimum lease payments for noncancelable operating leases were \$8,068 in 2003, \$4,937 in 2004, \$3,467 in 2005, \$2,424 in 2006, \$1,009 in 2007 and \$126 thereafter.

NOTE M — CONTINGENCIES

The Company, like other manufacturers, is subject from time to time to a variety of civil and administrative proceedings arising in the ordinary course of business. Such claims and litigation include, without limitation, product liability claims and health, safety and environmental claims, some of which relate to cases alleging asbestos and manganese induced illnesses. The Company believes it has meritorious defenses to these claims and intends to contest such suits vigorously. All costs associated with these claims, including defense and settlements, have been immaterial to the Company’s consolidated financial statements. Based on the Company’s historical experience in litigating these claims, including a significant number of dismissals, summary judgments and defense verdicts in many cases and immaterial settlement amounts, as well as the Company’s current assessment of the underlying merits of the claims and applicable insurance, the Company believes resolution of these claims and proceedings, individually or in the aggregate, will not have a material adverse impact upon the Company’s consolidated financial statements.

The Company has provided a guarantee on a loan for a joint venture. The Company believes the likelihood is remote that material payment will be required under this arrangement. The Company’s estimated maximum exposure under this arrangement at December 31, 2002 was approximately \$2,000.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE N — QUARTERLY FINANCIAL DATA (UNAUDITED)

2002	Mar 31	Jun 30	Sep 30	Dec 31
Net sales	\$248,343	\$259,715	\$247,486	\$238,533
Gross profit	74,048	81,565	73,217	71,195
Income before income taxes and the cumulative effect of a change in accounting principle	13,321	26,324	24,442	23,856
Net income (loss)	(27,123)	19,713	18,328	18,357
Basic earnings per share before the cumulative effect of a change in accounting principle	\$ 0.25	\$ 0.47	\$ 0.43	\$ 0.44
Cumulative effect of a change in accounting principle, net of tax	(0.89)	—	—	—
Basic earnings (loss) per share	\$ (0.64)	\$ 0.47	\$ 0.43	\$ 0.44
Diluted earnings per share before the cumulative effect of a change in accounting principle	\$ 0.25	\$ 0.46	\$ 0.43	\$ 0.43
Cumulative effect of a change in accounting principle, net of tax	(0.88)	—	—	—
Diluted earnings (loss) per share	\$ (0.63)	\$ 0.46	\$ 0.43	\$ 0.43
2001	Mar 31	Jun 30	Sep 30	Dec 31
Net sales	\$252,623	\$249,346	\$239,915	\$236,993
Gross profit	83,433	79,992	73,185	70,716
Income before income taxes	31,444	31,162	24,831	28,604
Net income	21,980	21,782	19,193	20,634
Basic earnings per share	\$ 0.52	\$ 0.51	\$ 0.45	\$ 0.49
Diluted earnings per share	\$ 0.52	\$ 0.51	\$ 0.45	\$ 0.48

The quarter ended March 31, 2002, includes a net pre-tax charge of \$10,468 (\$7,045 after-tax) relating to the Company’s rationalization program (see Note E) and a pre-tax charge for the cumulative effect of an accounting change of \$38,307 (\$37,607 after-tax) (see Note A).

The quarter ended December 31, 2001, includes a net pre-tax gain of \$1,943 (\$1,263 after-tax) related to a \$3,087 gain (\$2,007 after-tax) on the sale of property, partially offset by a charge of \$1,144 (\$744 after-tax) relating to severance and redundancy costs in Europe.

The quarterly earnings per share (EPS) amounts are each calculated independently. Therefore, the sum of the quarterly EPS amounts may not equal the annual totals.